

RESIDUAL MINIMISATION FOR TRANSPORT-BASED OPTIMAL CONTROL

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ABSTRACT. We consider an optimal control problem governed by a stationary linear transport equation in two or three spatial dimensions, motivated by photon transport in radiotherapy treatment planning. The control is an inflow boundary source, and the observation operator may track either the angular flux or its angular integral, the latter serving as a simplified dose surrogate. The associated optimality system consists of a transport–adjoint pair with angular dependence, coupled through a boundary optimality condition.

We formulate a residual least-squares functional whose minimisers coincide with solutions of the Karush–Kuhn–Tucker system and show that the functional controls the L^2 errors in the state and adjoint and the flux-weighted L^2 error in the control. This yields a continuum-level a posteriori estimate without recourse to discretisation. An additional unweighted optimality residual certifies the control in the norm of the objective. The estimate holds for every admissible approximation and accounts for the different boundary measures in the control penalty and transport traces, without a positive lower bound on $|\omega \cdot n|$. We then parameterise the state, adjoint and control using neural networks and minimise over a neural network space using appropriate quadrature. The Monte Carlo formulation replaces each residual integral by a weighted sample mean, whose quadrature error enters the a posteriori estimate.

We prove convergence under explicit assumptions on approximation, quadrature and optimisation and derive bounds on the total error in terms of approximation, quadrature and optimisation error. Numerical examples in two and three dimensions illustrate the residual–error relation and the flexibility of the residual framework for field-tracking control problems, including regular manufactured solutions.

1. INTRODUCTION

Optimal control problems governed by transport equations arise in a range of applied settings, from radiation therapy and shielding design to optical engineering and neutron flux regulation. In such problems, the control is typically applied at the inflow boundary of the domain, and the system response propagates along characteristics with angular structure and attenuation [FHS10].

This work is motivated by photon-based radiotherapy treatment planning, where an incident fluence is chosen to shape the radiation field within the patient [Bor06; FHS10]. Related transport-control questions arise in proton therapy, where energy loss and biological response also enter the model [Ash+25]. Here we isolate the boundary-control structure using a linear Boltzmann-type equation with pure attenuation. The control enters through the inflow boundary, and we consider either angular-flux tracking or tracking of its angular integral, the latter providing a simplified dose surrogate. The target-shape computations in Section 7 use angular-flux tracking and report the angular integral as an additional spatial diagnostic. Scattering, energy dependence, nonnegative delivery constraints and more complete dose models are left outside the present formulation.

We formulate a residual least-squares functional that encodes the full KKT system for the transport-based control problem. The residual functional $\mathcal{J}_b$ is quadratic, non-negative and vanishes only at the solution. We show that $\mathcal{J}_b$ provides a rigorous a posteriori estimate at the continuum level: its value controls the error in all components of the solution triple (u, z, g) . This enables residual-based convergence analysis and error certification independently of discretisation. The control error is measured in the transport boundary norm, with weight $|\omega \cdot n|$. The distinction between this norm and the unweighted control penalty is essential: the normal velocity enters the boundary optimality condition and may vanish at grazing directions. We establish stability without dividing by this velocity or excluding nearly grazing rays.

The unknowns are approximated using smooth feedforward neural networks and trained to minimise a quadrature approximation of $\mathcal{J}_b$ using gradient-based methods. The networks provide an approximation class on position-angle space, and automatic differentiation supplies the directional derivatives in the residual. The analysis starts from the continuum optimality system, so the stability estimate also applies to other approximations for which these residuals are well-defined. We analyse the resulting approximate residual

minimisers under both deterministic and Monte Carlo quadrature and prove convergence under explicit approximation, quadrature and optimisation hypotheses. Numerical experiments examine the relation between residual and error and demonstrate the feasibility of the approach in two and three dimensions.

Residual minimisation of full optimality systems is established in least-squares finite element methods [BG06]. Adaptive regularisation for elliptic control is studied in [PP25]. For transport equations, stable Petrov–Galerkin and least-squares formulations relate suitable residual norms to approximation errors [Dah+12; LZ20]. For the present control problem, the state and adjoint estimates must be combined with the boundary optimality equation. The relevant stability question therefore concerns the coupled system, including the error in its unknown inflow data.

Neural network approaches to PDE-constrained optimisation include [MN23; Lu+21]. In particular, joint neural approximation of the state, adjoint and control through optimality conditions is already used in [Bar+26]. The recent KKT-based method of [Zha+26] combines adversarial sampling with scaling to address sensitivity to the regularisation parameter. Deep Uzawa methods treat PDE and boundary constraints through multipliers [MPP24; MPP25], while the related stationary transport method [Mak+25] enforces prescribed inflow data. Here the inflow is itself the control, and its optimality condition couples the boundary trace of the adjoint to the regularisation term. This coupling must be retained when relating the residual to errors in all three unknowns.

The main contribution is a global residual-stability estimate for the transport boundary-control KKT system. It accounts simultaneously for the unweighted control penalty and the flux-weighted transport traces, without excluding nearly grazing directions. For each fixed $\alpha > 0$, the continuum residual controls the state and adjoint errors in L^2 and the control error in the transport boundary norm, independently of the particular approximation method. Strengthening only the optimality residual also yields control in the unweighted norm appearing in the objective and hence a bound on reduced-objective suboptimality.

This stability estimate separates the numerical error into approximation, quadrature and optimisation contributions. For Monte Carlo quadrature we distinguish estimates for a fixed approximation from uniform estimates over a bounded training class, since a network selected using the sampled loss is itself sample-dependent. Deterministic quadrature fits the same residual-error framework. The numerical experiments test the coupled state–adjoint–control approximation in two and three dimensions, compare target-control examples with a characteristic reference, and separately verify the fixed-field Monte Carlo quadrature rate.

The rest of the paper is organized as follows. Section 2 introduces the control problem and establishes well-posedness of the transport equations and their coupled optimality system. Section 3 characterises the residual minimisers and proves the global stability estimate. Section 4 describes the neural network parameterisation, quadrature loss and minimisation algorithm. Section 5 relates the discrete loss to the solution error through Monte Carlo and deterministic quadrature estimates. Approximation and convergence are considered in Section 6. The numerical results are presented in Section 7, followed by conclusions in Section 8.

2. PROBLEM SETUP

We study an optimal control problem arising in photon transport for radiotherapy treatment planning. The aim is to determine an incident flux that delivers a prescribed dose to a target region within tissue, modelled as a bounded, convex Lipschitz domain $\Omega \subset \mathbb{R}^d$, with $d = 2$ or 3 . The state of a photon is described by its spatial position $\mathbf{x} \in \Omega$ and direction $\boldsymbol{\omega} \in \mathbb{S}^{d-1}$. The flux $u = u(\mathbf{x}, \boldsymbol{\omega})$ represents the photon intensity at the corresponding phase-space point $(\mathbf{x}, \boldsymbol{\omega})$.

The full radiative transport model incorporates absorption, scattering and external sources, but to clarify the control structure we first consider the simplified setting without scattering or internal sources. In this case, the transport of photons is governed by a pure attenuation equation:

$$(1) \quad \boldsymbol{\omega} \cdot \nabla_{\mathbf{x}} u + \sigma(\mathbf{x})u = 0 \quad \text{in } \Omega \times \mathbb{S}^{d-1},$$

with inflow boundary condition

$$(2) \quad u|_{\Gamma^-} = g.$$

Here $\sigma \in L^\infty(\Omega)$ is nonnegative, and the inflow and outflow boundaries are

$$\Gamma^\pm = \{(\mathbf{x}, \boldsymbol{\omega}) \in \partial\Omega \times \mathbb{S}^{d-1} : \pm \boldsymbol{\omega} \cdot \mathbf{n}(\mathbf{x}) > 0\}.$$

The control variable is the inflow flux g , prescribed on Γ^- , and the forward field u solves (1)-(2). For treatment planning, we aim to deliver dose to a region $T \subset \Omega$. In the present model, the angular integral

$$D(u)(x) = \int_{\mathbb{S}^{d-1}} u(x, \omega) d\omega$$

provides a simplified dose surrogate. We may therefore match either the angular flux itself or its angular integral to a prescribed target. We write $\mathcal{O}u$ for the chosen quantity: $\mathcal{O} = I$ for angular-flux tracking and $\mathcal{O} = D$ for dose tracking, with $Y = L^2(\Omega \times \mathbb{S}^{d-1})$ and $Y = L^2(\Omega)$, respectively.

We consider the constrained minimisation problem

$$(3) \quad \min_{(u, g)} \frac{1}{2} \|\mathcal{O}u - X_T\|_Y^2 + \frac{\alpha}{2} \|g\|_{L^2(\Gamma^-)}^2 \quad \text{subject to (1), (2)}.$$

This is a linear-quadratic optimal control problem with PDE constraint and *boundary control*, where $\alpha > 0$ balances fidelity to the target and control cost. The target X_T belongs to Y . For angular-flux tracking it may depend on both x and ω ; a spatial target is understood to be constant in the angular variable. The analysis below applies to any bounded linear $\mathcal{O} : L^2(\Omega \times \mathbb{S}^{d-1}) \rightarrow Y$. The minimisation is over controls without a sign constraint. Quadratic dose-tracking objectives for kinetic treatment planning are also studied in [BFH12].

2.1. Transport graph space and boundary trace. We write $Q = \Omega \times \mathbb{S}^{d-1}$, $H = L^2(Q)$ and $b(x, \omega) = |\omega \cdot n(x)|$. The control cost in (3) uses the unweighted boundary space $U = L^2(\Gamma^-; dS_x d\omega)$, whereas integration by parts for transport uses

$$G_{\pm} = L^2(\Gamma^{\pm}; b dS_x d\omega).$$

The space U describes the control penalty, $G_{\pm}$ describe the transport boundary traces, and the space W below describes the state and adjoint. These are different roles. In particular,

$$\|h\|_{G_-}^2 = \int_{\Gamma^-} b|h|^2 dS_x d\omega \leq \int_{\Gamma^-} |h|^2 dS_x d\omega = \|h\|_U^2.$$

Thus $U \hookrightarrow G_-$ continuously. The converse bound fails near grazing directions. In the control objective we penalise the incident intensity with respect to boundary area and solid angle, thus the flux factor b instead enters the transport identity. This choice of control penalty is retained throughout. All angular integrals use surface measure on $\mathbb{S}^{d-1}$. If $v \in H$ and $\omega \cdot \nabla_x v \in H$, then along almost every characteristic the restriction of v has an H^1 representative and therefore well-defined endpoint values. We denote these characteristic endpoint values by γ^-v and γ^+v whenever the resulting boundary functions belong to G_- and G_+ , respectively.

We define

$$(4) \quad W = \{v \in H : \omega \cdot \nabla_x v \in H, \gamma^-v \in G_-, \gamma^+v \in G_+\},$$

with norm

$$\|v\|_W^2 = \|v\|_H^2 + \|\omega \cdot \nabla_x v\|_H^2 + \|\gamma^-v\|_{G_-}^2 + \|\gamma^+v\|_{G_+}^2.$$

Thus the existence of finite global transport traces is part of the definition of W . The transport graph norm alone need not control both boundary traces; the final two terms in $\|\cdot\|_W$ make this requirement explicit. This distinction also occurs in the transport framework of [Dah+12].

2.2. Lemma (Green identity for the transport operator). *The trace maps $\gamma^{\pm} : W \rightarrow G_{\pm}$ are linear and continuous, with*

$$\|\gamma^{\pm}v\|_{G_{\pm}} \leq \|v\|_W \quad (v \in W).$$

Let $T = \omega \cdot \nabla_x + \sigma$ and $T^* = -\omega \cdot \nabla_x + \sigma$. For $v, w \in W$,

$$(5) \quad (Tv, w)_H - (v, T^*w)_H = (\gamma^+v, \gamma^+w)_{G_+} - (\gamma^-v, \gamma^-w)_{G_-}.$$

Proof. The trace bounds follow from the definition of $\|\cdot\|_W$. Along almost every characteristic, a function and its directional derivative belong to L^2 , so its restriction belongs to H^1 on that interval. One-dimensional integration by parts therefore applies. For each fixed ω , orthogonal projection of the inflow and outflow boundaries onto $\omega^{\perp}$ has Jacobian $b = |\omega \cdot n|$. Integrating the one-dimensional identity over these projected

characteristics and over ω gives (5). The volume and boundary products are integrable by Cauchy–Schwarz and the definition of W . $\square$

For completeness, we record the transport estimates used below. In the present convex geometry they follow directly from the characteristic formula and require only nonnegative attenuation.

2.3. Theorem (Well-posedness of the transport equation). *Let $L = \text{diam}(\Omega)$. For $f \in H$ and $h \in G_-$, the problem $Tv = f$, $\gamma^- v = h$, has a unique solution $v \in W$, and*

$$(6) \quad \|v\|_H \leq L\|f\|_H + \sqrt{L}\|h\|_{G_-}, \quad \|\gamma^+ v\|_{G_+} \leq \sqrt{L}\|f\|_H + \|h\|_{G_-}.$$

*For $f \in H$ and $k \in G_+$, the adjoint problem $T^*w = f$, $\gamma^+ w = k$, has a unique solution $w \in W$, with*

$$(7) \quad \|w\|_H \leq L\|f\|_H + \sqrt{L}\|k\|_{G_+}, \quad \|\gamma^- w\|_{G_-} \leq \sqrt{L}\|f\|_H + \|k\|_{G_+}.$$

Proof. Let x_- be the inflow endpoint of a characteristic and let $0 < t < \ell(x_-, \omega) \leq L$ denote distance along it. Define

$$(8) \quad \begin{aligned} v(x_- + t\omega, \omega) = & \exp\left(-\int_0^t \sigma(x_- + s\omega) ds\right) h(x_-, \omega) \\ & + \int_0^t \exp\left(-\int_s^t \sigma(x_- + r\omega) dr\right) f(x_- + s\omega, \omega) ds. \end{aligned}$$

The characteristic change of variables is

$$\int_Q \varphi(x, \omega) dx d\omega = \int_{\Gamma^-} \int_0^{\ell(x_-, \omega)} \varphi(x_- + t\omega, \omega) dt b(x_-, \omega) dS_{x_-} d\omega.$$

Indeed, for each direction the orthogonal projection used in Lemma 2.2 parametrises the straight characteristics; convexity gives one interval on each ray. The same projection shows that the endpoint map from inflow to outflow preserves $b dS_x d\omega$. Since $\sigma \geq 0$, the attenuation factors in (8) are at most one. For the homogeneous term, integration along intervals of length at most L gives the bounds $\sqrt{L}\|h\|_{G_-}$ in H and $\|h\|_{G_-}$ at outflow. Cauchy–Schwarz gives the corresponding bounds $L\|f\|_H$ and $\sqrt{L}\|f\|_H$ for the source term. Thus (6) follows. The formula has inflow trace h , and $\omega \cdot \nabla_x v = f - \sigma v \in H$; the outflow estimate then gives $v \in W$. Uniqueness follows from the homogeneous ordinary differential equation on almost every characteristic. Reversing the direction gives the adjoint statement and (7). $\square$

2.4. Lagrangian formulation and optimality conditions. We use the standard Lagrangian approach to PDE-constrained optimisation [Hin+09]. We introduce the Lagrangian associated with (3) by incorporating the transport equation through a dual variable $z \in W$:

$$(9) \quad \mathcal{L}(u, g, z) = \frac{1}{2} \|\mathcal{O}u - X_T\|_Y^2 + \frac{\alpha}{2} \|g\|_U^2 - (Tu, z)_H.$$

The inflow condition is retained in the admissible pairs (u, g) . Thus their variations satisfy $\gamma^- \delta u = \delta g$. Using (5), the derivative in these directions is

$$\begin{aligned} D_{(u, g)} \mathcal{L}[\delta u, \delta g] = & (\mathcal{O}^*(\mathcal{O}u - X_T) - T^*z, \delta u)_H - (\gamma^+ z, \gamma^+ \delta u)_{G_+} \\ & + (\alpha g + b\gamma^- z, \delta g)_U. \end{aligned}$$

Stationarity therefore yields the KKT system

$$(10) \quad \begin{aligned} T^*z^* &= \mathcal{O}^*(\mathcal{O}u^* - X_T), & \gamma^+ z^* &= 0, \\ \alpha g^* + b\gamma^- z^* &= 0, \\ Tu^* &= 0, & \gamma^- u^* &= g^*. \end{aligned}$$

The boundary condition is equivalently $\alpha g^* = (\omega \cdot n)\gamma^- z^*$ on Γ^- . The factor b arises because the control penalty is unweighted, whereas the boundary term in Green’s identity is flux-weighted. For angular-flux tracking the adjoint source is $u^* - X_T$; for dose tracking it is $D(u^*) - X_T$, regarded as a function constant in ω . Indeed, $D^*p(x, \omega) = p(x)$ and $\|D\| \leq |\mathbb{S}^{d-1}|^{1/2}$.

2.5. Theorem (Well-posedness of the optimality system). *For every $X_T \in Y$ and $\alpha > 0$, problem (3) has a unique minimiser. Its state, adjoint and control form the unique solution $(u^*, z^*, g^*) \in W \times W \times U$ of (10). Moreover,*

$$(11) \quad \|g^*\|_U \leq \alpha^{-1/2} \|X_T\|_Y, \quad \|u^*\|_H \leq \sqrt{L/\alpha} \|X_T\|_Y, \quad \|z^*\|_H \leq L \|\mathcal{O}\| \|X_T\|_Y.$$

Proof. Let Sh denote the solution of $T(Sh) = 0$ with $\gamma^-(Sh) = h$. Theorem 2.3 gives $\|Sh\|_H \leq \sqrt{L} \|h\|_{G_-}$. Since $b \leq 1$, this estimate also holds with $\|h\|_U$ on the right. The reduced problem is to minimise

$$j(g) = \frac{1}{2} \|\mathcal{O}Sg - X_T\|_Y^2 + \frac{\alpha}{2} \|g\|_U^2.$$

Its stationarity equation is

$$\alpha(g, h)_U + (\mathcal{O}Sg, \mathcal{O}Sh)_Y = (X_T, \mathcal{O}Sh)_Y \quad (h \in U).$$

The bilinear form on the left is bounded and coercive on U with coercivity constant α . Lax–Milgram gives a unique stationary point g^* , which minimises j by strict convexity. Set $u^* = Sg^*$ and let z^* solve the adjoint equation with zero outflow data. Green’s identity gives

$$(\mathcal{O}u^* - X_T, \mathcal{O}Sh)_Y = (h, \gamma^- z^*)_{G_-} = (h, b\gamma^- z^*)_U.$$

Here $b\gamma^- z^* \in U$, since $\|bp\|_U \leq \|p\|_{G_-}$. The stationarity equation is therefore the boundary optimality condition in (10). Conversely, that system implies stationarity of j . Finally, $j(g^*) \leq j(0)$ implies $\sqrt{\alpha} \|g^*\|_U \leq \|X_T\|_Y$ and $\|\mathcal{O}u^* - X_T\|_Y \leq \|X_T\|_Y$. The transport estimates give (11). $\square$

3. RESIDUAL STABILITY

Following the least-squares principle [BG09], we measure all equations of the optimality system through their residuals. The optimality system may also be characterised by the residual functional

$$(12) \quad \mathcal{J}_b(u, z, g) = \frac{1}{2} (\|Tu\|_H^2 + \|T^*z - \mathcal{O}^*(\mathcal{O}u - X_T)\|_H^2 + \|\alpha g + b\gamma^- z\|_{G_-}^2 + \|\gamma^- u - g\|_{G_-}^2 + \|\gamma^+ z\|_{G_+}^2).$$

Although the original control problem is posed with $g \in U$, the weighted residual $\mathcal{J}_b$ is well defined on the larger space $W \times W \times G_-$. Since $U \hookrightarrow G_-$, this enlarges the residual domain without changing the zero-residual solution: as shown below, vanishing of the boundary optimality residual forces $g = -\alpha^{-1} b\gamma^- z \in U$.

Each term measures the violation of one interior or boundary equation. This formulation is convenient for neural network approximation, because the state, adjoint and control can be approximated together without imposing the constraints strongly. The boundary residuals use the transport measure, and the control error below is measured in the same norm.

3.1. Lemma (Characterisation of residual minimisers). *The solution (u^*, z^*, g^*) of (10) is the unique global minimiser of $\mathcal{J}_b$ over $W \times W \times G_-$, and its minimum value is zero. In particular,*

$$\mathcal{J}_b(u, z, g) = 0 \iff (u, z, g) = (u^*, z^*, g^*).$$

Proof. The KKT solution makes all five residuals vanish. Conversely, if $\mathcal{J}_b(u, z, g) = 0$, each squared residual norm is zero. The boundary optimality equation gives $g = -\alpha^{-1} b\gamma^- z \in U$, since $\|b\gamma^- z\|_U \leq \|\gamma^- z\|_{G_-}$. Thus $(u, z, g) \in W \times W \times U$ satisfies (10), and uniqueness follows from Theorem 2.5. Nonnegativity of $\mathcal{J}_b$ completes the proof. $\square$

3.2. Theorem (Global residual stability). *Under the hypotheses of Theorem 2.5, there is a constant $C = C(\alpha, L, \|\mathcal{O}\|) > 0$ such that every $(u, z, g) \in W \times W \times G_-$ satisfies*

$$(13) \quad \|u - u^*\|_H^2 + \|z - z^*\|_H^2 + \|g - g^*\|_{G_-}^2 \leq C \mathcal{J}_b(u, z, g).$$

An explicit choice is $C = C_{\text{stab}}$ in (35) below.

Proof. Write $e_u = u - u^*$, $e_z = z - z^*$ and $e_g = g - g^*$, and denote the five residuals in (12), in order, by $r_1, \dots, r_5$. Then

$$Te_u = r_1, \quad T^*e_z = \mathcal{O}^*\mathcal{O}e_u + r_2, \quad \alpha e_g + b\gamma^-e_z = r_3, \quad \gamma^-e_u = e_g + r_4, \quad \gamma^+e_z = r_5.$$

Let S denote the state lift used above, and let y solve $Ty = r_1$, $\gamma^-y = r_4$. Then $e_u = Se_g + y$. Let η solve

$$T^*\eta = \mathcal{O}^*\mathcal{O}y + r_2, \quad \gamma^+\eta = r_5.$$

The difference $w = e_z - \eta$ satisfies

$$T^*w = \mathcal{O}^*\mathcal{O}Se_g, \quad \gamma^+w = 0.$$

The state lift $S : G_- \rightarrow H$ is bounded, with $\|S\| \leq \sqrt{L}$ by Theorem 2.3. Hence

$$A := S^*\mathcal{O}^*\mathcal{O}S : G_- \rightarrow G_-$$

is bounded, self-adjoint and nonnegative, and

$$\|A\| \leq L\|\mathcal{O}\|^2.$$

For every $h \in G_-$, Green's identity gives

$$(h, \gamma^-w)_{G_-} = (Sh, \mathcal{O}^*\mathcal{O}Se_g)_H = (h, Ae_g)_{G_-},$$

so that $\gamma^-w = Ae_g$.

Let $P : G_- \rightarrow G_-$ denote multiplication by b . Since $0 \leq b \leq 1$, P is a bounded, self-adjoint, nonnegative contraction, and $P^{1/2}$ is multiplication by $\sqrt{b}$. The boundary equation therefore becomes

$$(14) \quad (\alpha I + PA)e_g = r_3 - P\gamma^-\eta.$$

Set

$$B = P^{1/2}AP^{1/2}.$$

Since A is self-adjoint and nonnegative, so is B . Thus

$$((\alpha I + B)v, v)_{G_-} \geq \alpha\|v\|_{G_-}^2,$$

and consequently $\alpha I + B$ is invertible with

$$\|(\alpha I + B)^{-1}\| \leq \alpha^{-1}.$$

Using the resolvent identity with $X = P^{1/2}$ and $Y = P^{1/2}A$,

$$(\alpha I + XY)^{-1} = \alpha^{-1}[I - X(\alpha I + YX)^{-1}Y],$$

we obtain

$$(15) \quad (\alpha I + PA)^{-1} = \alpha^{-1}[I - P^{1/2}(\alpha I + P^{1/2}AP^{1/2})^{-1}P^{1/2}A].$$

In particular, using $\|P^{1/2}\| \leq 1$,

$$\|(\alpha I + PA)^{-1}\| \leq \alpha^{-1}\left(1 + \frac{\|A\|}{\alpha}\right) \leq \kappa := \alpha^{-1}\left(1 + \frac{L\|\mathcal{O}\|^2}{\alpha}\right).$$

Writing $R_i = \|r_i\|$ in its respective residual space, the transport estimates imply

$$\begin{aligned} \|y\|_H &\leq LR_1 + \sqrt{L}R_4, \\ \|\gamma^-\eta\|_{G_-} &\leq \sqrt{L}(\|\mathcal{O}\|^2\|y\|_H + R_2) + R_5, \\ \|e_g\|_{G_-} &\leq \kappa(R_3 + \|\gamma^-\eta\|_{G_-}), \\ \|e_u\|_H &\leq \sqrt{L}\|e_g\|_{G_-} + \|y\|_H, \\ \|e_z\|_H &\leq L(\|\mathcal{O}\|^2\|e_u\|_H + R_2) + \sqrt{L}R_5. \end{aligned}$$

Combining these estimates and using $\sum_i R_i^2 = 2\mathcal{J}_b(u, z, g)$ proves (13). An explicit choice is $C_{\text{stab}} = 2\|\mathbf{M}\|_{\mathbb{F}}^2$, where $\mathbf{M}$ collects these coefficients as specified in Appendix A. For angular-flux tracking, the characteristic structure gives the sharper choice $\kappa = \alpha^{-1}$, and the dependence on α is examined there. $\square$

3.3. Remark (Boundary norms and loss weights). *If the three boundary residuals in (12) are measured in unweighted L^2 , denote the resulting functional by $\mathcal{J}_0$. Then $\mathcal{J}_b \leq \mathcal{J}_0$. In addition, the control equation and the adjoint trace estimate give*

$$\|e_g\|_U \leq \alpha^{-1} [\|r_3\|_U + \sqrt{L}(\|\mathcal{O}\|^2 \|e_u\|_H + \|r_2\|_H) + \|r_5\|_{G_+}].$$

Consequently, $\mathcal{J}_0$ also controls the unweighted control error whenever these residuals are finite. Multiplying the five squared residual norms by fixed positive weights λ_i preserves the stability estimate, with its constant multiplied by $1/\min_i \lambda_i$. This includes fixed normalisations by the volume and boundary measures.

3.4. Corollary (Control accuracy in the objective norm). *Under the hypotheses of Theorem 2.5, let $(u, z, g) \in W \times W \times U$ and define*

$$(16) \quad \mathcal{J}_U(u, z, g) = \frac{1}{2} (\|r_1\|_H^2 + \|r_2\|_H^2 + \|r_3\|_U^2 + \|r_4\|_{G_-}^2 + \|r_5\|_{G_+}^2).$$

There is an explicit $C_U = C_U(\alpha, L, \|\mathcal{O}\|)$, given in (36), such that

$$(17) \quad \|u - u^*\|_H^2 + \|z - z^*\|_H^2 + \|g - g^*\|_U^2 \leq C_U \mathcal{J}_U(u, z, g).$$

Moreover, writing $e_g = g - g^$, the reduced objective satisfies*

$$(18) \quad \begin{aligned} j(g) - j(g^*) &= \frac{1}{2} \|\mathcal{O} S e_g\|_Y^2 + \frac{\alpha}{2} \|e_g\|_U^2 \\ &\leq \frac{\alpha + L \|\mathcal{O}\|^2}{2} C_U \mathcal{J}_U(u, z, g). \end{aligned}$$

The constant C_U is of order α^{-2} for fixed L and $\|\mathcal{O}\|$ as $\alpha \downarrow 0$.

Proof. Multiplication by b maps G_- into U with norm at most one, so $r_3 = \alpha g + b\gamma^- z \in U$. Use $e_u = S e_g + y$ and $e_z = w + \eta$ from the proof of Theorem 3.2. Testing $\alpha e_g + b\gamma^- w = r_3 - b\gamma^- \eta$ against e_g in U , Green's identity gives

$$\alpha \|e_g\|_U^2 + \|\mathcal{O} S e_g\|_Y^2 = (r_3 - b\gamma^- \eta, e_g)_U, \quad \|e_g\|_U \leq \alpha^{-1} (\|r_3\|_U + \|\gamma^- \eta\|_{G_-}).$$

The same transport estimates for y , η , e_u and e_z , with $\|e_g\|_{G_-} \leq \|e_g\|_U$, prove (17). Their coefficients give the explicit constant in the appendix. For (18), expand $j(g)$ about g^* . The cross term vanishes by stationarity, and $\|S e_g\|_H \leq \sqrt{L} \|e_g\|_U$ gives the upper bound. $\square$

Only the optimality residual needs the stronger measure; no additional unweighted traces of u or z are required. Since $b \leq 1$,

$$(19) \quad \mathcal{J}_b \leq \mathcal{J}_U \leq \mathcal{J}_0$$

wherever $\mathcal{J}_0$ is finite. Thus the unweighted boundary loss also controls the error in U . A field trained with $\mathcal{J}_b$ may instead be assessed afterwards using $\mathcal{J}_U$. Weighted residual convergence alone does not imply convergence in U , as the grazing construction in Appendix A.4 shows. The objective $j(g)$ uses the feasible state Sg ; for fixed $\alpha > 0$, its convergence to $j(g^*)$ is equivalent to $g \rightarrow g^*$ in U by (18) and $\|S e_g\|_H \leq \sqrt{L} \|e_g\|_U$.

3.5. Remark (Characteristic reference for angular-flux tracking). *For $\mathcal{O} = I$, the characteristic formula also gives an explicit reference solution. At an inflow point (y, ω) , set*

$$a_y(t) = \exp \left(- \int_0^t \sigma(y + s\omega) ds \right), \quad 0 < t < \ell(y, \omega).$$

Then $u(y + t\omega, \omega) = a_y(t)g(y, \omega)$, and the reduced objective is the integral, with respect to $dS_y d\omega$, of

$$\frac{b(y, \omega)}{2} \int_0^{\ell(y, \omega)} |a_y(t)g - X_T(y + t\omega, \omega)|^2 dt + \frac{\alpha}{2} |g|^2.$$

Differentiation with respect to g gives

$$(20) \quad g^*(y, \omega) = \frac{b(y, \omega) \int_0^{\ell(y, \omega)} a_y(t) X_T(y + t\omega, \omega) dt}{\alpha + b(y, \omega) \int_0^{\ell(y, \omega)} a_y(t)^2 dt}.$$

The denominator is bounded below by α , and the adjoint is recovered from its transport equation with zero outflow data. For $X_T \geq 0$, the control in (20) is nonnegative. Dose tracking couples the angular directions through $D(u)$ and does not admit this separate minimisation along each ray.

4. NEURAL NETWORK APPROXIMATION AND RESIDUAL MINIMISATION

Neural residual minimisation [RPK19] has been used for radiative transfer in [MM21]. Here the network represents the spatial and angular dependence jointly; high-order space-angle-energy finite element approximations provide another approach to phase-space discretisation [Hou+24]. To approximate the optimality system (10), we represent the triple (u, z, g) using neural networks as smooth surrogate functions. These approximations are inserted directly into the residual functional $\mathcal{J}_b$ defined in (12), and trained via stochastic gradient descent to minimise the total loss. This constitutes a neural network-based residual minimisation scheme for the constrained transport control problem. Unlike traditional discretisations based on basis expansions, neural networks define a mesh-free function class that extends naturally to high-dimensional domains and automatically satisfies regularity conditions needed for the transport operator and boundary traces.

4.1. Neural network parameterisation of the KKT System. Let $V_\theta = (u_\theta, z_\theta, g_\theta)$ denote a feedforward network with three outputs, parametrised by weights $\theta \in \mathbb{R}^P$. The state and adjoint outputs are restricted to $\bar{\Omega} \times \mathbb{S}^{d-1}$, and the control output to Γ^- . Three separate scalar networks may also be used. We use C^∞ activation functions, such as $\tanh$, and write $\mathcal{A}_N$ for the resulting admissible class at resolution N .

By construction, the network functions are smooth in both spatial and angular variables and hence belong to the regularity class of the residual functional:

$$u_\theta, z_\theta \in C^\infty(\Omega \times \mathbb{S}^{d-1}), \quad (u_\theta, z_\theta, g_\theta) \in W \times W \times G_-.$$

The control g_θ is the restriction of a smooth ambient network to Γ^- . The membership in $W \times W \times G_-$ follows because the networks and their spatial derivatives are bounded on the compact set $\bar{\Omega} \times \mathbb{S}^{d-1}$. The directional derivatives $\omega \cdot \nabla_x u_\theta$ and $\omega \cdot \nabla_x z_\theta$ are therefore classically defined and computed by automatic differentiation. In two dimensions, using $(\cos(\vartheta), \sin(\vartheta))$ as the angular input imposes angular periodicity directly.

In fact, the boundedness of the control network and the finite unweighted measure of Γ^- give $g_\theta \in U$. Thus $V_\theta \in W \times W \times U$, and the stronger control estimate in Corollary 3.4 also applies when the optimality residual is measured in U .

4.2. Loss Functional and Quadrature Discretisation. Denote the five residuals in (12) by $r_1, \dots, r_5$. Let $\xi_j^\Omega \in Q$ and $\xi_j^\pm \in \Gamma^\pm$ be interior and boundary quadrature points, with positive weights w_j^Ω and $w_j^\pm$. Each ξ denotes a position-direction pair. The discrete loss is

$$(21) \quad \begin{aligned} \mathcal{J}_{b,h}(V) = & \frac{1}{2} \sum_{j=1}^{M_\Omega} w_j^\Omega |r_1(V; \xi_j^\Omega)|^2 + \frac{1}{2} \sum_{j=1}^{M_\Omega} w_j^\Omega |r_2(V; \xi_j^\Omega)|^2 \\ & + \frac{1}{2} \sum_{j=1}^{M_-} w_j^- |r_3(V; \xi_j^-)|^2 + \frac{1}{2} \sum_{j=1}^{M_-} w_j^- |r_4(V; \xi_j^-)|^2 + \frac{1}{2} \sum_{j=1}^{M_+} w_j^+ |r_5(V; \xi_j^+)|^2. \end{aligned}$$

The weights approximate $dx d\omega$ in the volume and $b dS_x d\omega$ on the boundary. For a product rule, the interior weights have the form $w_x w_\omega$, as in spatial and angular Gauss quadrature. We use $Q_{\Omega,h}$ and $Q_{\pm,h}$ below as shorthand for these weighted sums. For $\mathcal{J}_{U,h}$ and $\mathcal{J}_{0,h}$, the boundary quadrature weights are changed according to the corresponding residual measures.

Monte Carlo loss. Alternatively, the integrals can be approximated by sample means. Write

$$m_\Omega = |\Omega| |\mathbb{S}^{d-1}|, \quad m_\pm = \int_{\Gamma^\pm} b dS_x d\omega.$$

Draw M_Ω independent uniform samples in Q and, on each $\Gamma^\pm$, draw $M_\pm$ independent samples with probability measure $m_\pm^{-1} b dS_x d\omega$. Setting $w_j^\Omega = m_\Omega/M_\Omega$ and $w_j^\pm = m_\pm/M_\pm$ in (21) gives

$$(22) \quad \begin{aligned} \mathcal{J}_{b,M}(V) = & \frac{m_\Omega}{2M_\Omega} \sum_{j=1}^{M_\Omega} (|r_1(V; \xi_j^\Omega)|^2 + |r_2(V; \xi_j^\Omega)|^2) \\ & + \frac{m_-}{2M_-} \sum_{j=1}^{M_-} (|r_3(V; \xi_j^-)|^2 + |r_4(V; \xi_j^-)|^2) + \frac{m_+}{2M_+} \sum_{j=1}^{M_+} |r_5(V; \xi_j^+)|^2, \end{aligned}$$

where $M = (M_\Omega, M_-, M_+)$. The measure factors in (22) ensure that, for every fixed admissible V independent of the samples,

$$\mathbb{E}[\mathcal{J}_{b,M}(V)] = \mathcal{J}_b(V).$$

The same analysis applies to fixed positive rescalings of the residual terms, with the corresponding continuum functional and a modified stability constant.

4.3. Training Algorithm. The state, adjoint and control are trained jointly by minimising $\mathcal{J}_h(V_\theta)$. We collect all trainable weights in θ ; these may belong to one network with three outputs or to three separate networks. For Monte Carlo quadrature, the samples are drawn before training and held fixed during minimisation of $\mathcal{J}_{b,M}$. Deterministic rules are treated in the same way. We first use Adam [KB15] and, where specified, continue with L-BFGS [LN89]. The procedure is summarised below.

Here $\mathcal{J}_h$ denotes the quadrature approximation of the chosen residual $\mathcal{J}_b$, $\mathcal{J}_U$ or $\mathcal{J}_0$. Algorithm 1 applies to all three choices by using the corresponding boundary weights described above. The optimisation does not project the parameters onto a prescribed bounded set; the relation to the uniform sampling estimate is discussed after Proposition 5.6.

Algorithm 1 NN-based residual minimisation

Require: Initial parameters θ^0 , quadrature points and weights (sampled once for Monte Carlo), Adam learning rate η , and budgets N_A and N_B for Adam and L-BFGS

- 1: Define $\ell(\theta) = \mathcal{J}_h(V_\theta)$ by (21), using (22) for Monte Carlo samples
- 2: **for** $k = 0, \dots, N_A + N_B - 1$ **do**
- 3: Evaluate $u_\theta, z_\theta, g_\theta$ at the interior and boundary quadrature points
- 4: Compute $\omega \cdot \nabla_x u_\theta$ and $\omega \cdot \nabla_x z_\theta$ by automatic differentiation
- 5: Form the state, adjoint, optimality, inflow and outflow residuals $r_1, \dots, r_5$
- 6: Assemble $\ell(\theta^k)$ by the weighted sums in (21)
- 7: Compute gradient $\nabla_\theta \ell(\theta^k)$ by automatic differentiation through the residuals
- 8: Update all network parameters together:

$$\theta^{k+1} \leftarrow \begin{cases} \text{Adam}_\eta(\theta^k, \nabla_\theta \ell(\theta^k)), & k < N_A, \\ \text{LBFGS}(\theta^k; \ell, \nabla_\theta \ell), & k \geq N_A. \end{cases}$$

- 9: Record the evaluated loss and residual contributions, identifying the parameter iterate used
 - 10: **end for**
 - 11: **return** $(u_{\hat{\theta}}, z_{\hat{\theta}}, g_{\hat{\theta}})$, where $\hat{\theta} = \theta^{N_A + N_B}$, and the residual history
-

The Adam update retains its moment estimates between iterations. At the change of optimiser, the L-BFGS history is initialised. Each L-BFGS outer step may contain several inner iterations and line-search evaluations; the network values, spatial derivatives, residuals and parameter gradients are recomputed at each trial parameter vector. Thus a recorded L-BFGS step is not equivalent in cost to an Adam step. Setting $N_B = 0$ gives training with Adam alone.

For dose tracking, the adjoint residual uses $D_h u_\theta(x) = \sum_i w_i u_\theta(x, \omega_i)$ in place of the angular integral. Its parameter derivative therefore couples all directions at each spatial point. We also denote this implemented loss by $\mathcal{J}_{b,h}$; its discrepancy from $\mathcal{J}_b$ includes the inner angular approximation, as detailed in Section 5. If the calculation is divided into spatial chunks, each chunk retains the complete angular rule and its weighted gradient contributions are accumulated before updating the parameters.

All conditions, including the inflow data and the boundary optimality law, enter through the residual functional. The continuum stability estimate applies to every returned network. To infer an error bound from the discrete loss, the quadrature error must also be controlled. For Monte Carlo quadrature, this is a sampling error, which motivates the probabilistic estimates in the next section.

5. QUADRATURE ERROR AND A POSTERIORI CONTROL

The computed loss is a quadrature approximation to the continuum residual. We now determine how this approximation enters the control of the state, adjoint and control errors. Residual stability first gives a deterministic estimate in terms of the discrete loss and its quadrature discrepancy. For Monte Carlo quadrature, we bound this discrepancy in mean square and with high probability, distinguishing fixed fields from fields selected during training. We then consider deterministic rules and the inner angular quadrature for dose tracking.

5.1. A posteriori residual control. For $V = (u, z, g)$ and the unique optimality-system solution V^* , write $E = H \times H \times G_-$ and

$$\|V - V^*\|_E^2 = \|u - u^*\|_H^2 + \|z - z^*\|_H^2 + \|g - g^*\|_{G_-}^2.$$

The global stability estimate gives an a posteriori bound for every admissible approximation, independently of its construction.

5.2. Corollary (A posteriori residual control). *Every $V \in W \times W \times G_-$ satisfies*

$$\|V - V^*\|_E^2 \leq C_{\text{stab}} \mathcal{J}_b(V),$$

where C_{stab} depends only on α , L and $\|\mathcal{O}\|$. In particular, $\mathcal{J}_b(V_N) \rightarrow 0$ implies $V_N \rightarrow V^*$ in E . If $|\mathcal{J}_b(V) - \mathcal{J}_{b,h}(V)| \leq q_h(V)$, then

$$\|V - V^*\|_E^2 \leq C_{\text{stab}} (\mathcal{J}_{b,h}(V) + q_h(V)).$$

Proof. Theorem 3.2 gives

$$\|V - V^*\|_E^2 \leq C_{\text{stab}} \mathcal{J}_b(V) \leq C_{\text{stab}} [\mathcal{J}_{b,h}(V) + |\mathcal{J}_b(V) - \mathcal{J}_{b,h}(V)|].$$

The assertions follow by applying the first inequality to V_N and the second to the stated quadrature bound. $\square$

Thus the continuum residual controls the solution error without requiring the approximation to lie near the solution. A small discrete loss has the same interpretation once its quadrature error is bounded. The relation between stability, residual loss and quadrature error is also central to the general analyses of neural residual minimisation, e.g. in [MM23; SZK23]. In particular, for the Monte Carlo loss,

$$(23) \quad \|V - V^*\|_E^2 \leq C_{\text{stab}} [\mathcal{J}_{b,M}(V) + |\mathcal{J}_b(V) - \mathcal{J}_{b,M}(V)|].$$

This inequality holds for every realisation of the samples. It separates the loss reduced during training from the sampling error that must be controlled to infer accuracy of the solution.

For the control-cost norm, if $V \in W \times W \times U$ and $|\mathcal{J}_U(V) - \mathcal{J}_{U,h}(V)| \leq q_{U,h}(V)$, Corollary 3.4 similarly gives

$$(24) \quad \|u - u^*\|_H^2 + \|z - z^*\|_H^2 + \|g - g^*\|_U^2 \leq C_U [\mathcal{J}_{U,h}(V) + q_{U,h}(V)].$$

Multiplication by $(\alpha + L\|\mathcal{O}\|^2)/2$ also bounds $j(g) - j(g^*)$. By (19), the same argument applies to $\mathcal{J}_0$ with its own quadrature discrepancy. These are reliability bounds; the constants can be conservative, as discussed in Appendix A. A loss evaluated on training nodes alone is a diagnostic, not a continuum certificate.

5.3. Monte Carlo quadrature. We begin with the sample-mean approximation introduced in (22). For a fixed field its error is governed by the moments of the residual densities. When the field is chosen by minimising the sampled loss, uniform control over the neural class allows the same quadrature argument to be used.

The sample-mean loss (22) approximates five residual integrals. To cover both boundary sampling choices, write

$$\mathcal{J}_b(V) = \sum_{k=1}^K \int_{D_k} F_k(V; y) d\mu_k(y), \quad K = 5,$$

where $F_k = |r_k|^2/2$, and μ_k is the volume or flux-weighted boundary measure. Fixed positive loss weights may be included in F_k . For a sampling probability measure ν_k with $\mu_k \ll \nu_k$, set

$$R_k(V; y) = F_k(V; y) \frac{d\mu_k}{d\nu_k}(y), \quad \mathcal{J}_{b,M}(V) = \sum_{k=1}^K \frac{1}{M_k} \sum_{j=1}^{M_k} R_k(V; Y_{k,j}).$$

For each k , the samples $Y_{k,j}$ are independent with law ν_k . In (22), $M_1 = M_2 = M_\Omega$, $M_3 = M_4 = M_-$ and $M_5 = M_+$; the two volume terms share their samples, as do the two inflow terms. The arguments below allow this sharing between residual terms.

5.4. Lemma (Mean-square quadrature error). *Fix V independently of the samples and suppose $R_k(V; \cdot) \in L^2(\nu_k)$ for $k = 1, \dots, K$. Then $\mathbb{E}[\mathcal{J}_{b,M}(V)] = \mathcal{J}_b(V)$ and*

$$(25) \quad \left(\mathbb{E}[|\mathcal{J}_{b,M}(V) - \mathcal{J}_b(V)|^2] \right)^{1/2} \leq \sum_{k=1}^K \left(\frac{\text{Var}_{\nu_k}(R_k(V; \cdot))}{M_k} \right)^{1/2}.$$

Proof. Set $I_k = \int_{D_k} F_k(V; y) d\mu_k(y)$ and $X_{k,j} = R_k(V; Y_{k,j}) - I_k$. For each fixed k , independence and centering give

$$\mathbb{E}[X_{k,j}] = 0, \quad \mathbb{E}[X_{k,j}X_{k,\ell}] = 0 \quad (j \neq \ell), \quad \mathbb{E}[X_{k,j}^2] = \text{Var}_{\nu_k}(R_k(V; \cdot)).$$

Consequently, with $\Delta_k = M_k^{-1} \sum_j X_{k,j}$,

$$\mathbb{E}[\Delta_k] = 0, \quad \mathbb{E}[|\Delta_k|^2] = \frac{1}{M_k^2} \sum_{j=1}^{M_k} \mathbb{E}[X_{k,j}^2] = \frac{\text{Var}_{\nu_k}(R_k(V; \cdot))}{M_k}.$$

Since $\mathcal{J}_{b,M}(V) - \mathcal{J}_b(V) = \sum_k \Delta_k$, unbiasedness follows by linearity of expectation. Minkowski's inequality yields

$$\left(\mathbb{E} \left[\left| \sum_k \Delta_k \right|^2 \right] \right)^{1/2} \leq \sum_k (\mathbb{E}[|\Delta_k|^2])^{1/2},$$

which proves (25) without requiring independence between different residual terms. $\square$

This is the classical variance argument for Monte Carlo integration [Caf98], applied separately to the volume and boundary residuals.

With a common sample count $M_k = M$, the root-mean-square quadrature error is therefore $O(M^{-1/2})$ for fixed fields. This requires finite second moments of the squared residual densities; well-posedness in L^2 alone does not supply them. Bounded densities give the following high-probability estimate.

5.5. Proposition (Monte Carlo quadrature control). *Fix $V \in W \times W \times G_-$ independently of the samples and let $0 < \delta < 1$. Suppose finite bounds B_k satisfy $0 \leq R_k(V; y) \leq B_k$ for ν_k -almost every y . Then, with probability at least $1 - \delta$,*

$$(26) \quad |\mathcal{J}_{b,M}(V) - \mathcal{J}_b(V)| \leq \sum_{k=1}^K B_k \sqrt{\frac{\log(2K/\delta)}{2M_k}}.$$

On the same event,

$$(27) \quad \|V - V^*\|_E^2 \leq C_{\text{stab}} \left[\mathcal{J}_{b,M}(V) + \sum_{k=1}^K B_k \sqrt{\frac{\log(2K/\delta)}{2M_k}} \right].$$

Proof. With Δ_k as in the preceding proof, Hoeffding's inequality [Hoe63] gives, for $B_k > 0$ and $t > 0$,

$$\mathbb{P}(|\Delta_k| > t) \leq 2 \exp\left(-\frac{2M_k t^2}{B_k^2}\right).$$

Choose $t_k = B_k \sqrt{\log(2K/\delta)/(2M_k)}$. If $B_k = 0$, then $\Delta_k = t_k = 0$ almost surely. Hence

$$\mathbb{P}(\exists k : |\Delta_k| > t_k) \leq \sum_{k: B_k > 0} 2 \exp\left(-\frac{2M_k t_k^2}{B_k^2}\right) \leq \delta.$$

On the complementary event,

$$|\mathcal{J}_{b,M}(V) - \mathcal{J}_b(V)| = \left| \sum_k \Delta_k \right| \leq \sum_k t_k.$$

This proves (26); substitution into (23) gives (27). $\square$

Here B_k bounds the range of the sampled density. A variance bound suffices for (25), but not for this exponential concentration estimate. A field selected by minimising $\mathcal{J}_{b,M}$ depends on the samples. We therefore need a quadrature estimate that holds simultaneously throughout the admissible neural class.

Uniform control on the training class.

5.6. Proposition (Uniform quadrature control). *For a fixed architecture, let the admissible parameter set $\Theta_N \subset \mathbb{R}^{P_N}$ be nonempty and compact, and write $R_k(\theta; y) = R_k(V_\theta; y)$. Suppose, on a common set of full ν_k -measure, that for every $\theta, \vartheta \in \Theta_N$,*

$$0 \leq R_k(\theta; y) \leq B_k, \quad |R_k(\theta; y) - R_k(\vartheta; y)| \leq L_k \|\theta - \vartheta\|.$$

Define the uniform discrepancy by

$$\varepsilon_{quad}(N, M) = \sup_{\theta \in \Theta_N} |\mathcal{J}_b(V_\theta) - \mathcal{J}_{b,M}(V_\theta)|.$$

For $\eta > 0$, let $\mathcal{N}_N(\eta)$ be the cardinality of a finite η -net of Θ_N , with its centres in Θ_N . For $0 < \delta < 1$, with probability at least $1 - \delta$,

$$(28) \quad \varepsilon_{quad}(N, M) \leq 2\eta \sum_{k=1}^K L_k + \sum_{k=1}^K B_k \sqrt{\frac{\log(2K\mathcal{N}_N(\eta)/\delta)}{2M_k}}.$$

Proof. Let $\{\theta_\ell\}_{\ell=1}^{\mathcal{N}_N(\eta)}$ be the net and define

$$\Delta_k(\theta) = \frac{1}{M_k} \sum_{j=1}^{M_k} R_k(\theta; Y_{k,j}) - \int R_k(\theta; y) d\nu_k(y).$$

For $\|\theta - \theta_\ell\| \leq \eta$, the common Lipschitz bound implies, almost surely,

$$|\Delta_k(\theta) - \Delta_k(\theta_\ell)| \leq \frac{1}{M_k} \sum_{j=1}^{M_k} L_k \eta + \int L_k \eta d\nu_k = 2L_k \eta.$$

It follows that

$$\sup_{\theta \in \Theta_N} \left| \sum_k \Delta_k(\theta) \right| \leq 2\eta \sum_k L_k + \sum_k \max_\ell |\Delta_k(\theta_\ell)|.$$

Set $t_k = B_k \sqrt{\log(2K\mathcal{N}_N(\eta)/\delta)/(2M_k)}$. Applying Hoeffding's inequality at the net points gives

$$\mathbb{P}(\exists k, \ell : |\Delta_k(\theta_\ell)| > t_k) \leq \sum_{k: B_k > 0} \sum_{\ell=1}^{\mathcal{N}_N(\eta)} 2 \exp\left(-\frac{2M_k t_k^2}{B_k^2}\right) \leq \delta.$$

Terms with $B_k = 0$ vanish. On the complementary event, inserting these thresholds into the preceding supremum bound proves (28). $\square$

The event is uniform over Θ_N , so it applies to an approximation selected using those same samples, provided its parameters belong to Θ_N . The constants B_k , L_k and the covering number may grow with the architecture; sample sizes must increase accordingly. This is the distinction between quadrature control for a fixed approximation and quadrature control throughout training.

In particular, for a parameter $\hat{\theta} \in \Theta_N$ returned by minimising this sampled loss,

$$\|V_{\hat{\theta}} - V^*\|_E^2 \leq C_{\text{stab}}[\mathcal{J}_{b,M}(V_{\hat{\theta}}) + \varepsilon_{\text{quad}}(N, M)].$$

Together with (28), this gives the probabilistic counterpart of the discrete residual estimate for the trained approximation. It is this uniform quadrature control that enters the convergence theorem below.

Proposition 5.6 is a sufficient conditional result for bounded parameter classes. The implemented Adam/L-BFGS iterations are not projected onto a prescribed compact set Θ_N , so the hypotheses of the proposition are not asserted for the optimisation trajectories used in Section 7. Accordingly, Proposition 5.6 is used here as a convergence framework, rather than as a probabilistic certificate for the reported computations. For a fixed trained network, an independent-sample certificate may instead be obtained under the conditions described next.

Independent validation. A second use of Proposition 5.5 is to evaluate a trained approximation on fresh samples. Freeze $\hat{V}$ before drawing these samples. Conditional on training, its residual densities are fixed, so (27) holds with $V = \hat{V}$ and with the loss evaluated on the validation samples, provided the stated bounds B_k hold conditionally. The probability bound remains valid after averaging over training. This gives an a posteriori estimate without requiring uniform control over every network that training could have selected.

This is an available route to certification, rather than a validation procedure performed for the reported trained networks. It requires fresh independent samples and justified conditional bounds on the residual densities; reevaluation on a finer deterministic quadrature rule alone does not supply this probabilistic certificate.

The independent-sample estimates above concern independent position–direction pairs within each residual term. Reusing a direction at many spatial points, or combining shared spatial and angular sample sets, introduces dependence between the resulting pairs. Their product count is therefore not the number of independent samples in these estimates. The three-dimensional experiment uses such a sampling structure; its recorded training errors are not assigned a confidence bound by substituting that product count into (26).

For $\mathcal{J}_U$, the same measure-based analysis uses $\mu_3 = dS_x d\omega$ for the optimality term. The corresponding sampling weights and moment conditions are set out in Appendix A.5.

5.7. Deterministic quadrature.

5.8. Lemma (Polynomial quadrature consistency). *Let Q_h be a positive quadrature rule with nodes in D for a finite measure μ , exact on a polynomial space $\mathcal{P}_h$ containing constants. For a bounded integrand with specified pointwise values, write $\|F\|_{\infty,D} = \sup_{y \in D} |F(y)|$. Then*

$$(29) \quad \left| \int_D F d\mu - Q_h F \right| \leq 2\mu(D) \inf_{p \in \mathcal{P}_h} \|F - p\|_{\infty,D}.$$

Proof. Write $Q_h F = \sum_j w_j F(y_j)$. Positivity and exactness on constants give $w_j \geq 0$ and $\sum_j w_j = \mu(D)$. For every $p \in \mathcal{P}_h$, polynomial exactness therefore yields

$$\begin{aligned} \left| \int_D F d\mu - Q_h F \right| &= \left| \int_D (F - p) d\mu - \sum_j w_j (F - p)(y_j) \right| \\ &\leq \int_D |F - p| d\mu + \sum_j w_j |F - p|(y_j) \\ &\leq 2\mu(D) \|F - p\|_{\infty,D}. \end{aligned}$$

Taking the infimum over p proves the estimate. $\square$

The estimate applies to each residual integral separately. For a composite rule, applying it on each subinterval or integration cell D_ℓ , with local exactness space $\mathcal{P}_{h,\ell}$, gives

$$\left| \int_D F d\mu - Q_h F \right| \leq 2 \sum_\ell \mu(D_\ell) \inf_{p \in \mathcal{P}_{h,\ell}} \|F - p\|_{\infty, D_\ell}.$$

A q -point Gauss–Legendre rule is exact through degree $2q - 1$; its tensor product has this property in each coordinate. A spherical t -design is exact for spherical polynomials of degree at most t . Consequently, quadrature convergence follows from approximation of the complete residual densities, including the coefficients, target data and any geometric Jacobians. For a bound uniform over a neural class, the corresponding approximation estimates must also be uniform over that class. Analytic extension to a common complex neighbourhood with uniform bounds gives geometric polynomial-approximation estimates. Finite differentiability gives the algebraic rate permitted by the corresponding approximation estimate (see [Tre19, Chapters 7–8 and 19]) for these classical polynomial-approximation and quadrature arguments. Smooth activation functions alone do not ensure exponential quadrature convergence; discontinuous targets or coefficient interfaces require a separate piecewise analysis.

5.9. Angular quadrature for dose tracking. For dose tracking, quadrature also enters inside the adjoint residual through $D_h u$. Let $r_{2,h}$ denote the residual obtained by replacing Du by $D_h u$. Then

$$r_2 - r_{2,h} = D_h u - Du, \quad \|r_2 - r_{2,h}\|_H = |\mathbb{S}^{d-1}|^{1/2} \|Du - D_h u\|_{L^2(\Omega)},$$

because the difference is constant in the angular variable. Polarisation and the Cauchy–Schwarz inequality give

$$\begin{aligned} \frac{1}{2} \left| \|r_2\|_H^2 - \|r_{2,h}\|_H^2 \right| &= \frac{1}{2} |(r_2 - r_{2,h}, r_2 + r_{2,h})_H| \\ &\leq \frac{|\mathbb{S}^{d-1}|^{1/2}}{2} (\|r_2\|_H + \|r_{2,h}\|_H) \|Du - D_h u\|_{L^2(\Omega)}. \end{aligned}$$

This consistency term is added to the error in the outer quadrature. Accordingly, the discrepancy $|\mathcal{J}_b(V) - \mathcal{J}_{b,h}(V)|$ includes both approximations.

If the outer quadrature is Monte Carlo, the estimates above apply to the modified residual with a fixed inner angular rule independent of the outer samples. The displayed consistency term is then added to control the original residual. For a uniform training estimate it must also be bounded uniformly over the neural class. In particular, an unbiased estimate of Du does not by itself give an unbiased estimate of the squared adjoint residual.

6. NEURAL NETWORK APPROXIMATION AND CONVERGENCE

We now combine approximation in the transport space with the quadrature estimates. The first result gives sufficient conditions for the continuum residual to be approximated arbitrarily close by neural networks. The convergence estimate then separates this approximation error from quadrature and optimisation.

6.1. Approximation in the graph and trace norm. We next consider approximation of the KKT solution by neural networks. Since the residual contains transport derivatives and boundary traces, approximation in L^2 alone is not sufficient. We use the space

$$\mathbb{X} = W \times W \times G_-, \quad \|V\|_{\mathbb{X}}^2 = \|u\|_W^2 + \|z\|_W^2 + \|g\|_{G_-}^2,$$

where W carries the graph and trace norm introduced in Subsection 2.1. Simultaneous approximation of a function and its derivatives by neural networks was established in [HSW90]. For the tanh networks used here, we use the constructive approximation result from [DLM21].

6.2. Proposition (Neural network approximation). *Suppose V^* belongs to the $\mathbb{X}$ -closure of triples obtained by restricting smooth ambient functions to the phase domain and its inflow boundary. Let the nested neural classes $\mathcal{A}_N$ contain, eventually, every three-output tanh network with two hidden layers and finite width and parameters. Then*

$$\inf_{V \in \mathcal{A}_N} \|V - V^*\|_{\mathbb{X}} \longrightarrow 0, \quad \inf_{V \in \mathcal{A}_N} \mathcal{J}_b(V) \leq C \inf_{V \in \mathcal{A}_N} \|V - V^*\|_{\mathbb{X}}^2 \longrightarrow 0.$$

A sufficient condition for the closure hypothesis is

$$u^*, z^* \in L^2(\mathbb{S}^{d-1}; H^1(\Omega)), \quad g^* \in G_-.$$

Thus no angular derivatives are required for this sufficient condition. It is an additional regularity assumption, rather than a consequence of well-posedness for arbitrary L^2 data.

Proof. Fix $\varepsilon > 0$. By the closure hypothesis, choose a smooth ambient triple V_ε such that $\|V_\varepsilon - V^*\|_{\mathfrak{X}} < \varepsilon/2$. After multiplication by a smooth cutoff, its components may be taken smooth on a compact box B containing $\bar{\Omega} \times \mathbb{S}^{d-1}$. For any smooth ambient triple $V = (v_1, v_2, v_3)$, finiteness of the volume and boundary measures and $|\omega| = 1$ imply

$$\|V\|_{\mathfrak{X}} \leq C_B \left(\sum_{i=1}^3 \|v_i\|_{C^1(B)}^2 \right)^{1/2}.$$

Indeed, the right-hand side controls the two volume norms, the spatial directional derivatives and every weighted trace norm. By [DLM21, Theorem 5.1], each component of V_ε admits a two-hidden-layer tanh approximation in $C^1(B)$. Placing the three scalar networks in parallel gives a three-output network V_{θ_ε} such that

$$\left(\sum_{i=1}^3 \|(V_{\theta_\varepsilon} - V_\varepsilon)_i\|_{C^1(B)}^2 \right)^{1/2} < \frac{\varepsilon}{2C_B}.$$

This finite network belongs to $\mathcal{A}_N$ for all sufficiently large N . Hence

$$\inf_{V \in \mathcal{A}_N} \|V - V^*\|_{\mathfrak{X}} \leq \|V_{\theta_\varepsilon} - V_\varepsilon\|_{\mathfrak{X}} + \|V_\varepsilon - V^*\|_{\mathfrak{X}} < \varepsilon.$$

For completeness, consider the stated sufficient condition. Since Ω is Lipschitz, there is a bounded extension operator from $H^1(\Omega)$ into $H^1(\mathbb{R}^d)$, and extension followed by mollification shows that restrictions of functions smooth on a neighbourhood of $\bar{\Omega}$ are dense in $H^1(\Omega)$. Together with the density of smooth functions in $L^2(\mathbb{S}^{d-1})$ and the density of finite-rank tensors in the Bochner space, it follows that finite sums

$$\sum_{j=1}^m a_j(\omega) v_j(x), \quad a_j \in C^\infty(\mathbb{S}^{d-1}),$$

with each v_j smooth on a neighbourhood of $\bar{\Omega}$, are dense in $L^2(\mathbb{S}^{d-1}; H^1(\Omega))$. Smooth functions on the sphere admit smooth extensions to a neighbourhood of $\mathbb{S}^{d-1}$, so these tensor sums are restrictions of smooth ambient functions.

For $w \in L^2(\mathbb{S}^{d-1}; H^1(\Omega))$, the spatial trace theorem and $b \leq 1$ give

$$\|\omega \cdot \nabla_x w\|_H \leq \|\nabla_x w\|_H, \quad \|\gamma^- w\|_{G_-}^2 + \|\gamma^+ w\|_{G_+}^2 \leq C \|w\|_{L^2(\mathbb{S}^{d-1}; H^1(\Omega))}^2.$$

Hence, if w_m is such a smooth tensor approximation,

$$\|w_m - w\|_W \leq C \|w_m - w\|_{L^2(\mathbb{S}^{d-1}; H^1(\Omega))} \longrightarrow 0.$$

For the control, let $K = \partial\Omega \times \mathbb{S}^{d-1}$ and extend g^* by zero from Γ^- to K . The measure $b dS_x d\omega$ is a finite Borel measure on the compact set K , so continuous functions on K are dense in the corresponding L^2 space. By the Stone–Weierstrass theorem, restrictions to K of polynomials in the ambient coordinates are uniformly dense in $C(K)$. Since such polynomials are smooth ambient functions, their restrictions to Γ^- approximate g^* in G_- . This proves the closure hypothesis.

Finally, write $e_u = u - u^*$, $e_z = z - z^*$ and $e_g = g - g^*$. Subtracting the zero residuals at V^* gives

$$\begin{aligned} \|r_1\|_H &\leq \|\omega \cdot \nabla_x e_u\|_H + \|\sigma\|_\infty \|e_u\|_H, \\ \|r_2\|_H &\leq \|\omega \cdot \nabla_x e_z\|_H + \|\sigma\|_\infty \|e_z\|_H + \|\mathcal{O}\|^2 \|e_u\|_H, \\ \|r_3\|_{G_-} &\leq \alpha \|e_g\|_{G_-} + \|\gamma^- e_z\|_{G_-}, \\ \|r_4\|_{G_-} &\leq \|\gamma^- e_u\|_{G_-} + \|e_g\|_{G_-}, \quad \|r_5\|_{G_+} = \|\gamma^+ e_z\|_{G_+}. \end{aligned}$$

Squaring and summing, and using the definition of $\mathbb{X}$, we obtain

$$2\mathcal{J}_b(V) = \sum_{k=1}^5 \|r_k\|^2 \leq C(\alpha, \|\sigma\|_\infty, \|\mathcal{O}\|) \|V - V^*\|_{\mathbb{X}}^2,$$

where each residual uses its corresponding norm. Taking the infimum over $\mathcal{A}_N$ completes the proof. $\square$

The same argument applies to any smooth activation for which approximation in C^1 is available. The result concerns increasing neural classes, allowing the width and parameter bounds to grow; it does not assert consistency of a single fixed architecture. Together with residual stability, it implies convergence of any sequence of continuum near-minimisers whose optimisation gaps tend to zero.

6.3. Convergence. We conclude by combining approximation, quadrature and optimisation. For each neural resolution N , choose a quadrature rule $h = h(N)$. For Monte Carlo quadrature, this amounts to choosing sample counts $M_k = M_k(N)$, and $\mathcal{J}_{b,h}$ denotes $\mathcal{J}_{b,M}$. For classes $\mathcal{A}_N = \{V_\theta : \theta \in \Theta_N\}$ satisfying the preceding compactness, boundedness and Lipschitz hypotheses, the uniform discrepancy is controlled by (28). The three contributions retain the notation $\varepsilon_{\text{approx}}(N)$, $\varepsilon_{\text{quad}}(N)$ and $\varepsilon_{\text{opt}}(N)$.

6.4. Theorem (Convergence under approximation, quadrature and optimisation). *Let $\mathcal{A}_N \subset \mathbb{X}$ be a nonempty admissible neural class and $\widehat{V}_N \in \mathcal{A}_N$ a computed approximation satisfying*

$$\mathcal{J}_{b,h}(\widehat{V}_N) \leq \inf_{V \in \mathcal{A}_N} \mathcal{J}_{b,h}(V) + \varepsilon_{\text{opt}}(N), \quad \varepsilon_{\text{opt}}(N) \geq 0.$$

Define

$$\begin{aligned} \varepsilon_{\text{approx}}(N) &:= \inf_{V \in \mathcal{A}_N} \mathcal{J}_b(V), \\ \varepsilon_{\text{quad}}(N) &:= \sup_{V \in \mathcal{A}_N} |\mathcal{J}_b(V) - \mathcal{J}_{b,h}(V)|. \end{aligned}$$

If $\varepsilon_{\text{quad}}(N)$ is finite, then

$$(30) \quad \|\widehat{V}_N - V^*\|_E^2 \leq C_{\text{stab}}(\varepsilon_{\text{approx}}(N) + 2\varepsilon_{\text{quad}}(N) + \varepsilon_{\text{opt}}(N)).$$

Consequently, if all three contributions tend to zero, the state, adjoint and control converge to the KKT solution in the error norm $\|\cdot\|_E$.

Proof. For $\varepsilon > 0$, choose $V_{N,\varepsilon} \in \mathcal{A}_N$ such that $\mathcal{J}_b(V_{N,\varepsilon}) \leq \varepsilon_{\text{approx}}(N) + \varepsilon$. Then

$$\begin{aligned} \mathcal{J}_b(\widehat{V}_N) &\leq \mathcal{J}_{b,h}(\widehat{V}_N) + \varepsilon_{\text{quad}}(N) \\ &\leq \mathcal{J}_{b,h}(V_{N,\varepsilon}) + \varepsilon_{\text{opt}}(N) + \varepsilon_{\text{quad}}(N) \\ &\leq \mathcal{J}_b(V_{N,\varepsilon}) + \varepsilon_{\text{opt}}(N) + 2\varepsilon_{\text{quad}}(N) \\ &\leq \varepsilon_{\text{approx}}(N) + \varepsilon + \varepsilon_{\text{opt}}(N) + 2\varepsilon_{\text{quad}}(N). \end{aligned}$$

Let $\varepsilon \downarrow 0$ and apply Theorem 3.2. No minimiser of the continuum or discrete loss over $\mathcal{A}_N$ needs to be attained. $\square$

The corresponding conclusion in U requires $\|r_3(\widehat{V}_N)\|_U \rightarrow 0$ in addition to the three vanishing error contributions above, with $\widehat{g}_N \in U$. Indeed,

$$\mathcal{J}_U(\widehat{V}_N) \leq \mathcal{J}_b(\widehat{V}_N) + \frac{1}{2} \|r_3(\widehat{V}_N)\|_U^2 \rightarrow 0,$$

so Corollary 3.4 gives $\widehat{g}_N \rightarrow g^*$ in U and $j(\widehat{g}_N) \rightarrow j(g^*)$. The added hypothesis concerns the continuum residual, i.e. its quadrature discrepancy must also be controlled. Smoothness of the network alone does not imply this convergence.

6.5. Remark (Approximation, quadrature and optimisation). *A related separation of errors is used in the analysis of deep Uzawa methods [Mak+25]. Here residual stability connects these three errors to the state, adjoint and control. Proposition 6.2 gives sufficient conditions for the approximation error to vanish, while Section 5 supplies deterministic and probabilistic quadrature bounds. The factor two in (30) reflects the use of quadrature at both the computed network and the continuum comparison function. The optimisation gap remains an assumption on the returned approximation; convexity of the residual in the fields does not*

imply convexity in the network parameters or convergence of a particular training algorithm. Its nonnegative empirical loss is an upper bound for this gap.

6.6. Corollary (Convergence for random quadrature). *Under the hypotheses of Theorem 6.4, suppose deterministic sequences $a_N, o_N, q_N \geq 0$ tend to zero, with $\varepsilon_{\text{approx}}(N) \leq a_N$, $\varepsilon_{\text{opt}}(N) \leq o_N$ almost surely, and*

$$\mathbb{P}(\varepsilon_{\text{quad}}(N) \leq q_N) \geq 1 - \delta_N, \quad 0 < \delta_N < 1.$$

If $\delta_N \rightarrow 0$, then $\widehat{V}_N \rightarrow V^$ in probability in E . If $\sum_N \delta_N < \infty$, the convergence is almost sure.*

Proof. On $A_N = \{\varepsilon_{\text{quad}}(N) \leq q_N\}$, the total-error estimate gives, almost surely,

$$\|\widehat{V}_N - V^*\|_E^2 \leq C_{\text{stab}}(a_N + 2q_N + o_N).$$

For any $\varepsilon > 0$, the right-hand side is at most ε^2 for all sufficiently large N . Thus

$$\mathbb{P}(\|\widehat{V}_N - V^*\|_E > \varepsilon) \leq \mathbb{P}(A_N^c) \leq \delta_N \rightarrow 0.$$

If $\sum_N \delta_N < \infty$, Borel–Cantelli gives $\mathbb{P}(A_N^c \text{ infinitely often}) = 0$. The deterministic bound then implies almost-sure convergence. Independence between refinement levels is not needed. $\square$

The estimate concerns the squared solution error. An additive quadrature contribution of order $M^{-1/2}$ therefore gives an $M^{-1/4}$ contribution to the solution norm, with the associated confidence and class-complexity factors. Obtaining an accuracy–cost bound would additionally require quantitative approximation rates and a bound on the cost of reducing the optimisation gap.

7. NUMERICAL RESULTS

The numerical experiments examine the approximation of the optimality system and the behaviour of the residual during training. The target-control examples are motivated by irradiation of a prescribed region: we consider circular and triangular targets in the square, followed by a localised target in the unit disc. These computations use angular-flux tracking, with $\mathcal{O} = I$ and adjoint source $u - X_T$, and we plot the angular integral $\phi_\theta = D(u_\theta)$ as a dose surrogate. Manufactured solutions in the disc and the three-dimensional ball allow comparison with the exact state, adjoint and control. Finally, a separate Monte Carlo experiment examines the quadrature error for fixed approximating fields.

The loss families used below are summarised by

Examples 7.1, 7.2	Examples 7.3, 7.4, 7.6	Fixed-field Monte Carlo study
$\mathcal{J}_{0,h}$	$\mathcal{J}_{b,h}$	$\mathcal{J}_{b,M}$

Each training loss may include the fixed positive normalisations specified in its example. The $\mathcal{J}_U$ diagnostic is evaluated after training in the manufactured disc example. Unless stated otherwise, the reported network errors are evaluated on the same quadrature points used during training and should therefore be read as discrete accuracy diagnostics rather than independent continuum certificates.

For the deterministic computations on $\Omega = [0, 1]^2$, composite three-point Gauss–Legendre rules on 30 subintervals in each spatial coordinate give 8100 interior nodes. The angular rule is also composite Gauss–Legendre, with the number of subintervals specified below. The angular integrals are evaluated by the weighted sum of the directional fields. The directional panels within each figure use a common colour scale.

7.1. Example 7.1: Circular target under directional irradiation. We consider $\Omega = [0, 1]^2$ with constant attenuation $\sigma = 0.1$. The target region is a disc $T = B_{0.1}(0.5, 0.5)$, and the desired field is $X_T = \chi_T$. We take $\alpha = 10^{-4}$ and use 75 angular nodes, obtained from a three-point rule on 25 subintervals of $[0, 2\pi]$. The selected computation uses a network with 12 hidden layers of width 64 and tanh activations, and records 50 000 training iterations.

The three boundary residuals are integrated with the unweighted measure $dS_x d\omega$. Thus this computation uses the boundary norms of $\mathcal{J}_0$, up to fixed positive loss normalisations, while the optimality equation remains $\alpha g + b\gamma^- z = 0$. By (19), this stronger choice of residual measure falls within the control estimate of Corollary 3.4.

Figure 1 shows two directional fluxes, their angular integral and the training loss. The elongated profiles follow straight characteristics through the circular target. Their location is determined by the projection

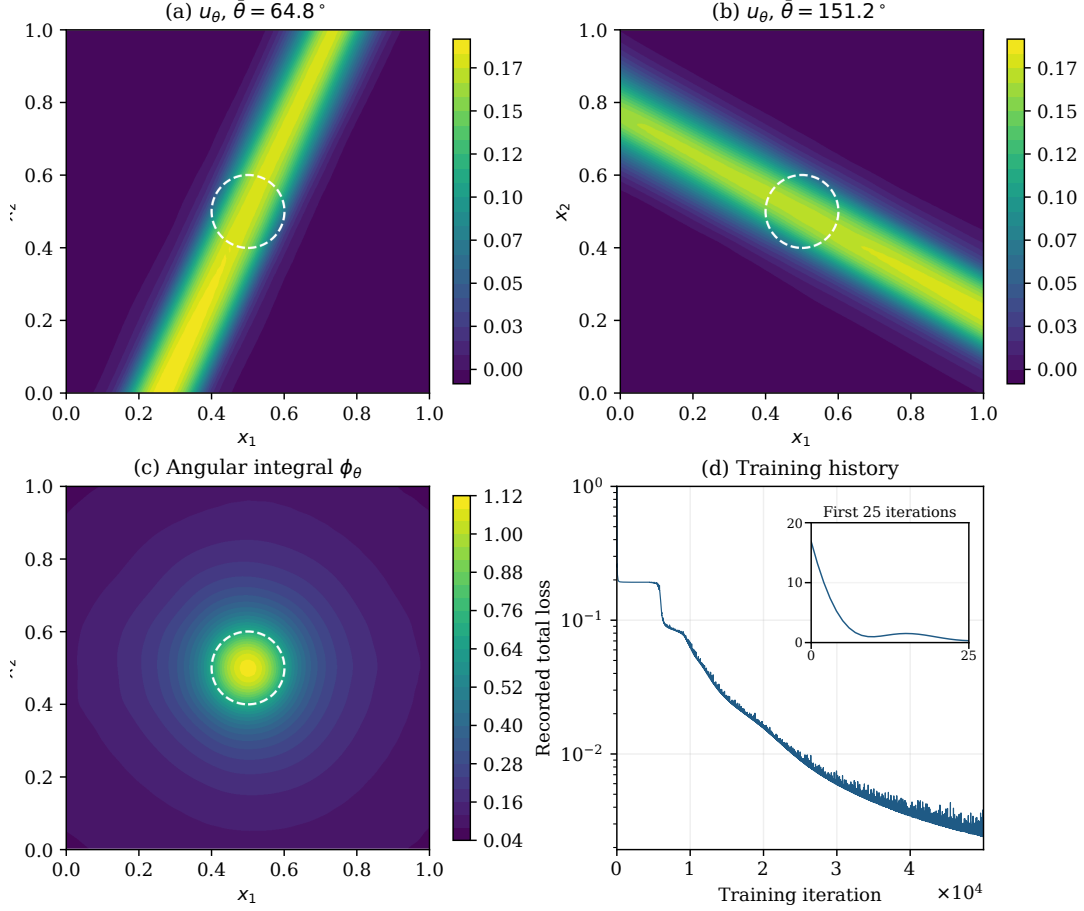


FIGURE 1. Circular target. Directional fields at 64.8° and 151.2° , angular integral over all 75 quadrature directions, and the recorded loss history. The dashed circle marks the target. The inset shows the first 25 iterations on a linear scale; the main loss axis focuses on values below 1. All panels are from the same computation.

of T onto the inflow boundary along each direction, while their transverse variation reflects the different lengths of intersection with the target. Formula (20) makes this dependence explicit: the optimal inflow is zero on rays that miss T , and its value on intersecting rays balances their attenuation-weighted overlap with T against the field energy and control penalty. The computed directional fields reproduce the resulting beam structure.

Integrating these fields over the 75 quadrature directions produces a maximum near the centre of the target and a gradual decrease away from it. A directional flux cannot be confined to T , since it propagates from the boundary and continues beyond the target along the same characteristic. The surrounding contribution in the angular integral is therefore part of the transport response. The loss decreases to 2.517×10^{-3} , with a slower decrease towards the end of training. The characteristic comparison below quantifies how accurately this computation recovers the optimal field.

7.2. Example 7.2: Triangular target and angular variation. We consider again $\Omega = [0, 1]^2$ and attenuation $\sigma = 0.1$, but replace the target by the triangle with vertices

$$\{(0.75, 0.75), (0.6, 0.25), (0.15, 0.5)\}.$$

We prescribe the tracking field $X_T = \chi_T$ and take $\alpha = 10^{-3}$. The angular rule has 45 nodes, from a three-point rule on 15 subintervals of $[0, 2\pi]$. As in the circular example, the network has 12 hidden layers of width 64, and training uses 50 000 iterations.

We retain the unweighted boundary residuals of the circular example. The comparison with the characteristic optimum therefore concerns the same unweighted-control formulation under this stronger residual measure.

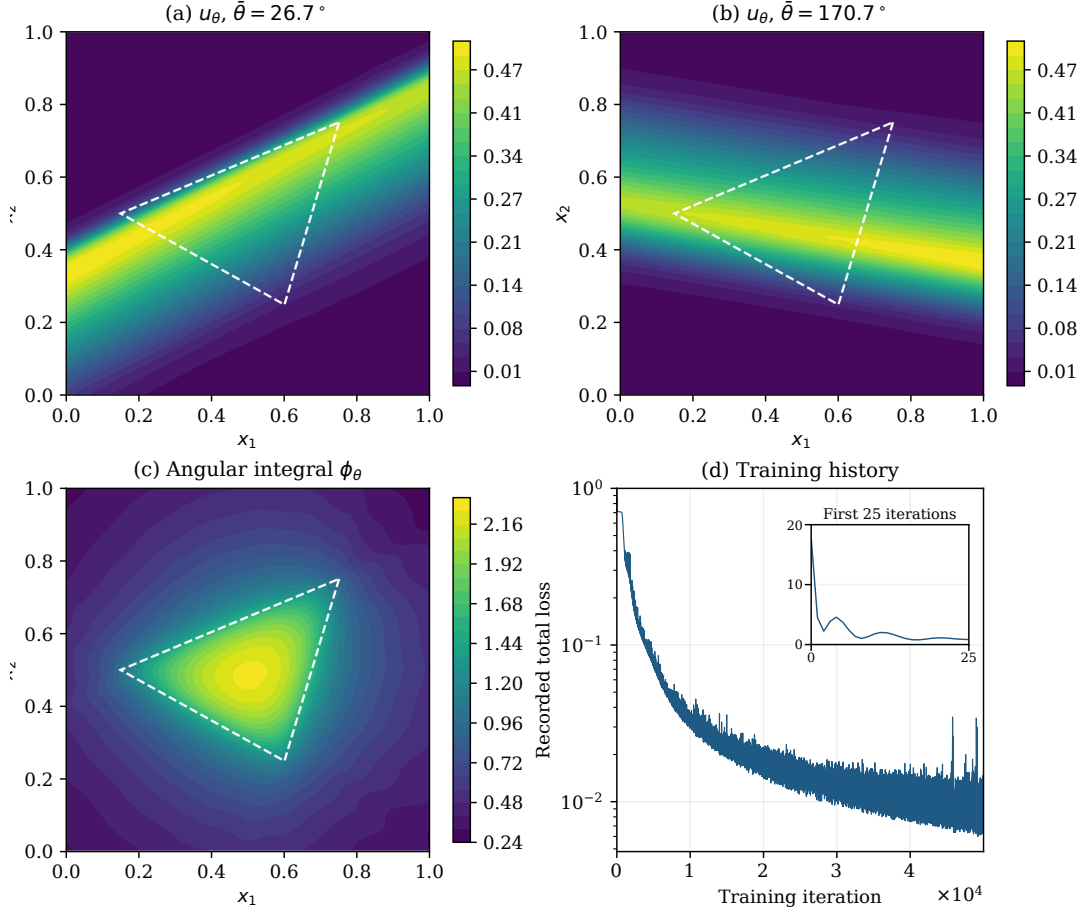


FIGURE 2. Triangular target. Directional fields at 26.7° and 170.7° , angular integral over all 45 quadrature directions, and the recorded loss history. The dashed triangle marks the target. The inset shows the first 25 iterations on a linear scale; the main loss axis focuses on values below 1. All panels are from the same computation.

The directional profiles in Figure 2 reflect the direction-dependent projections and chord lengths of the triangular target. By (20), these quantities determine the support and magnitude of the optimal inflow, so changing the incidence direction produces a corresponding change in the transported field. The angular integral retains the triangular geometry with the expected smoothing caused by transport along complete characteristics. The recorded loss decreases to 7.582×10^{-3} , although fluctuations remain visible near the end of training.

7.3. Example 7.3: Regular manufactured KKT solution in the unit disc. We now consider a manufactured solution in the unit disc, designed to satisfy the full KKT system exactly while remaining regular near grazing directions. This example provides the main quantitative benchmark for the residual minimisation scheme.

Let $\Omega = \{x \in \mathbb{R}^2 : |x| < 1\}$, let $\omega \in S^1$, and take $\sigma \equiv 1$. For $(x, \omega) \in \Omega \times S^1$ define

$$d = x \cdot \omega, \quad \Delta = \sqrt{(x \cdot \omega)^2 + 1 - |x|^2},$$

and

$$\tau_-(x, \omega) = d + \Delta, \quad \tau_+(x, \omega) = -d + \Delta.$$

Let

$$A(\Delta) = \begin{cases} \exp(-1/\Delta^2), & \Delta > 0, \\ 0, & \Delta = 0. \end{cases}$$

We prescribe

$$u_{\text{ex}}(x, \omega) = A(\Delta) \exp(-x \cdot \omega).$$

Since Δ is constant along characteristics, u_{ex} satisfies

$$\omega \cdot \nabla u_{\text{ex}} + u_{\text{ex}} = 0 \quad \text{in } \Omega \times S^1.$$

The corresponding inflow datum is

$$g_{\text{ex}}(x_{\text{in}}, \omega) = A(\Delta) e^\Delta,$$

where $x_{\text{in}} = x - \tau_-(x, \omega)\omega$.

Define

$$r(\Delta) = -\frac{\alpha A(\Delta) e^\Delta}{\Delta(1 - e^{-2\Delta})},$$

with the removable value $r(0) = 0$, and set

$$z_{\text{ex}}(x, \omega) = r(\Delta)(1 - e^{-\tau_+(x, \omega)}).$$

Finally define the manufactured tracking field by

$$X_T(x, \omega) = u_{\text{ex}}(x, \omega) - r(\Delta).$$

Then

$$-\omega \cdot \nabla z_{\text{ex}} + z_{\text{ex}} = u_{\text{ex}} - X_T \quad \text{in } \Omega \times S^1,$$

and $z_{\text{ex}} = 0$ on Γ^+ . Moreover, on Γ^- one has $\omega \cdot n(x_{\text{in}}) = -\Delta$ and $\tau_+(x_{\text{in}}, \omega) = 2\Delta$, hence

$$(\omega \cdot n)z_{\text{ex}} = (-\Delta)r(\Delta)(1 - e^{-2\Delta}) = \alpha A(\Delta) e^\Delta = \alpha g_{\text{ex}}.$$

Thus $(u_{\text{ex}}, z_{\text{ex}}, g_{\text{ex}})$ satisfies the full KKT system exactly.

The same construction applies in the unit ball in three dimensions. Indeed, $\omega \cdot \nabla d = 1$, $\omega \cdot \nabla \Delta = 0$ and $\omega \cdot \nabla \tau_+ = -1$ in either dimension. The factor $\exp(-1/\Delta^2)$ decays faster than every power of Δ , so all the apparent singularities at grazing incidence are removable. We set $\alpha = 1$ in the following computations. The factor $A(\Delta)$ suppresses rays close to grazing incidence, where the boundary optimality law would otherwise introduce small denominators. This gives a regular reference with a nontrivial state, adjoint and control, while retaining the coupling through the inflow boundary.

Discretisation and diagnostics. The manufactured-solution solvers represent (u, z, g) jointly by a network with three outputs, eight hidden layers of width 64, and tanh activations. The disc solver takes $(\cos(\vartheta), \sin(\vartheta), x_1, x_2)$ as input, while the ball solver takes $(\omega_1, \omega_2, \omega_3, x_1, x_2, x_3)$. The supplied implementation uses Xavier-uniform initialisation and single-precision arithmetic. The manufactured benchmarks test recovery of the whole optimality system, including the inflow control, rather than only the forward transport equation.

The recorded loss is a sum of normalised residual means,

$$(31) \quad \hat{L} = \frac{Q_{\Omega, h}(r_1^2 + r_2^2)}{Q_{\Omega, h}(1)} + \frac{Q_{-, h}(r_3^2 + r_4^2)}{Q_{-, h}(1)} + \frac{Q_{+, h}(r_5^2)}{Q_{+, h}(1)}.$$

All five residual multipliers equal one. We report relative errors

$$(e_u^{\text{rel}})^2 = \frac{Q_{\Omega, h}((u_\theta - u_{\text{ex}})^2)}{Q_{\Omega, h}(u_{\text{ex}}^2)}, \quad (e_z^{\text{rel}})^2 = \frac{Q_{\Omega, h}((z_\theta - z_{\text{ex}})^2)}{Q_{\Omega, h}(z_{\text{ex}}^2)}, \quad (e_g^{\text{rel}})^2 = \frac{Q_{-, h}((g_\theta - g_{\text{ex}})^2)}{Q_{-, h}(g_{\text{ex}}^2)}.$$

These diagnostics are evaluated on the training nodes. In particular, $\hat{L}$ differs from the unnormalised integral residual by its measure factors and the factor $1/2$.

In the disc, three-point Gauss-Legendre rules on 30 radial and 30 polar-angle subintervals give 8100 spatial nodes, including the radial Jacobian. Combining them with 30 angular nodes gives 243 000 interior

phase-space points. The boundary rule combines 900 circle nodes with the same directions, giving 13 500 nodes on each of Γ^- and Γ^+ .

The disc history contains 1 250 recorded steps. Its transition near step 500 is consistent with the supplied implementation’s initial Adam phase followed by L-BFGS. The implementation uses Adam learning rate 10^{-4} and L-BFGS learning rate 1, with a strong Wolfe line search and at most 20 inner iterations per outer step. Thus a recorded step is not a common unit of computational cost for the two optimisers.

Results in the disc. Table 1 reports the final relative and absolute errors, evaluated with the training quadrature. The absolute errors are measured in H for the state and adjoint and in G_- for the control; $\hat{J}_b$ denotes the corresponding integral residual.

Diagnostic	State	Adjoint	Control	Residual
Relative	$9.935 \cdot 10^{-4}$	$5.845 \cdot 10^{-4}$	$5.944 \cdot 10^{-4}$	$\hat{L} = 1.879 \cdot 10^{-6}$
Absolute	$1.6297 \cdot 10^{-3}$	$1.4126 \cdot 10^{-3}$	$1.3955 \cdot 10^{-3}$	$\hat{J}_b = 1.6609 \cdot 10^{-5}$

TABLE 1. Regular disc benchmark after 1 250 recorded steps. The control errors use the flux-weighted boundary measure. $\hat{J}_b$ is the quadrature approximation of (12).

Diagnostic	Transport-weighted G_-	Unweighted U
Absolute control error	1.3955×10^{-3}	2.4361×10^{-3}
Relative control error	5.9439×10^{-4}	1.0072×10^{-3}
Optimality residual norm	1.5147×10^{-3}	2.4801×10^{-3}

TABLE 2. Postprocessing the final saved predictions of the same disc computation. Each relative error uses the exact-control norm in its own measure. The optimality residual is $\alpha g_\theta + b z_\theta$ in both columns; only its integration measure changes.

Table 2 uses the original composite Gauss rule, with 900 boundary nodes and 30 directions, giving 13 500 inflow phase-space nodes. The unweighted control error is about 0.10%, compared with 0.059% in the transport norm. Thus the weighted diagnostic does not conceal a large unweighted error on these points. Recomputed weighted errors and the optimality residual mean square agree with the archived values to relative discrepancies below 10^{-6} . This is postprocessing of a computation trained with weighted boundary residuals; it is not a new training run or an independent-quadrature certificate.

The quadrature masses are $Q_{\Omega,h}(1) = 2\pi^2$ and $Q_{\pm,h}(1) \simeq 4\pi$. Restoring these measures in the five stored residual means gives

$$\widehat{E}_b^2 = E_u^2 + E_z^2 + E_g^2 \simeq 6.5989 \times 10^{-6}, \quad \widehat{E}_b^2 / \hat{J}_b \simeq 0.3973.$$

This observed squared-error/residual ratio is not an estimate of the global reliability constant. For this geometry and $\alpha = 1$, the raywise choice in Table 4 gives $C_{\text{stab}} = 712$, illustrating the conservatism of the global bound even though the observed residual and errors decrease together. Figure 3 shows the error histories and this residual–error comparison. The state, adjoint and control errors all decrease during training, with the final relative errors at or below 10^{-3} . This tests the simultaneous recovery of the transport equations and boundary optimality condition. The relevant comparison for Theorem 3.2 is between the sum of squared absolute errors and the integral residual. Restoring the measure factors is essential here: the normalised residual means in $\hat{L}$ weight the volume and boundary contributions differently. The decrease of $\widehat{E}_b^2$ together with $\hat{J}_b$, and their displayed ratio, are consistent with the stability estimate. These quantities use the same training quadrature, so this comparison does not determine the sharpness of its continuum constant.

The directional fields in Figure 4 also reflect the characteristic construction. Along each ray the state attenuates from its prescribed inflow value. The adjoint is negative and approaches zero at the outflow end, as follows from $r(\Delta) < 0$ and $\tau_+ \rightarrow 0$. Across rays, the factor $A(\Delta)$ gives a smooth decay towards grazing

incidence. These profiles complement the error histories by showing the distinct roles of the state and adjoint boundary conditions.

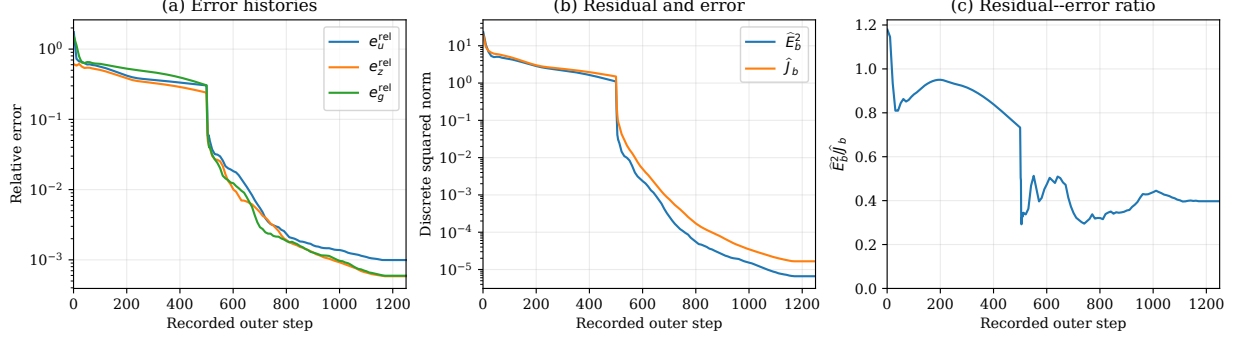


FIGURE 3. Regular disc benchmark. Relative-error histories, absolute squared error and integral residual, and their ratio. The residual is reconstructed with the quadrature masses and the factor $1/2$. All quantities use the training quadrature.

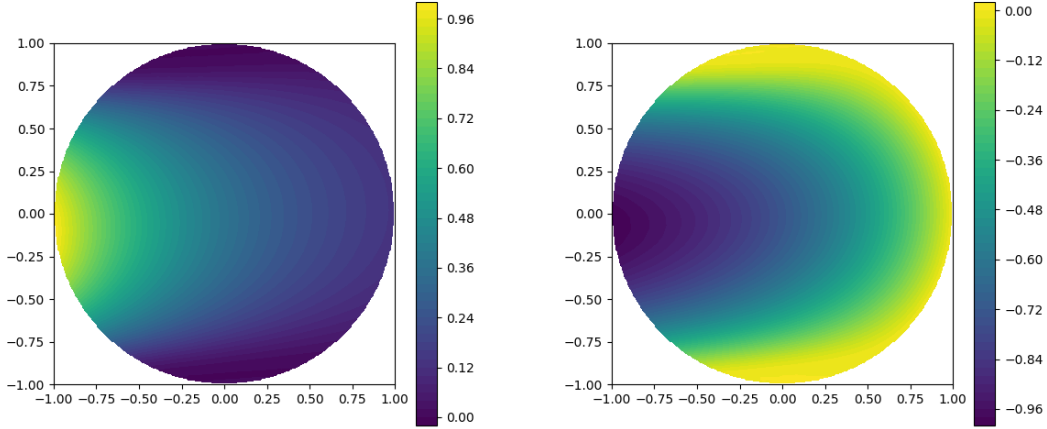


FIGURE 4. Regular disc benchmark. State (left) and adjoint (right) at the first angular quadrature node. The panels use linear spatial interpolation of the final saved predictions.

7.4. Example 7.4: Directional irradiation of a localised target. We next return to a radiotherapy-motivated target geometry, now in the unit disc. This remains an angular-flux tracking problem; the angular integral is shown as a spatial dose surrogate in order to illustrate the illumination produced by the optimised inflow.

Let Ω be the unit disc in $\mathbb{R}^2$ and set

$$X_T(x, \omega) = \chi_T(x), \quad T = \{x \in \Omega : |x| < 0.3\}.$$

We take $\sigma = 1$ and $\alpha = 10^{-4}$. As in the first two examples, the target is prescribed for the angular flux. We examine the resulting directional profiles together with their angular integral

$$\phi_\theta(x) = \sum_{i=1}^{45} w_i u_\theta(x, \omega_i), \quad \sum_{i=1}^{45} w_i = 2\pi.$$

This provides a simple model of irradiation of a localised region and allows us to examine the distribution of the dose surrogate both inside and outside the target.

We use 8100 spatial quadrature points and 45 directions, obtained from composite three-point Gauss–Legendre rules, and 270 boundary quadrature points. The network has twelve hidden layers with 64 neurons per layer and a hyperbolic tangent activation. Training uses 50000 Adam iterations with learning rate 5×10^{-4} . The recorded sum of the five squared residual integrals is 1.27×10^{-3} at the final iteration; this is twice the discrete approximation of $\mathcal{J}_b$, owing to the factor $1/2$ in (12).

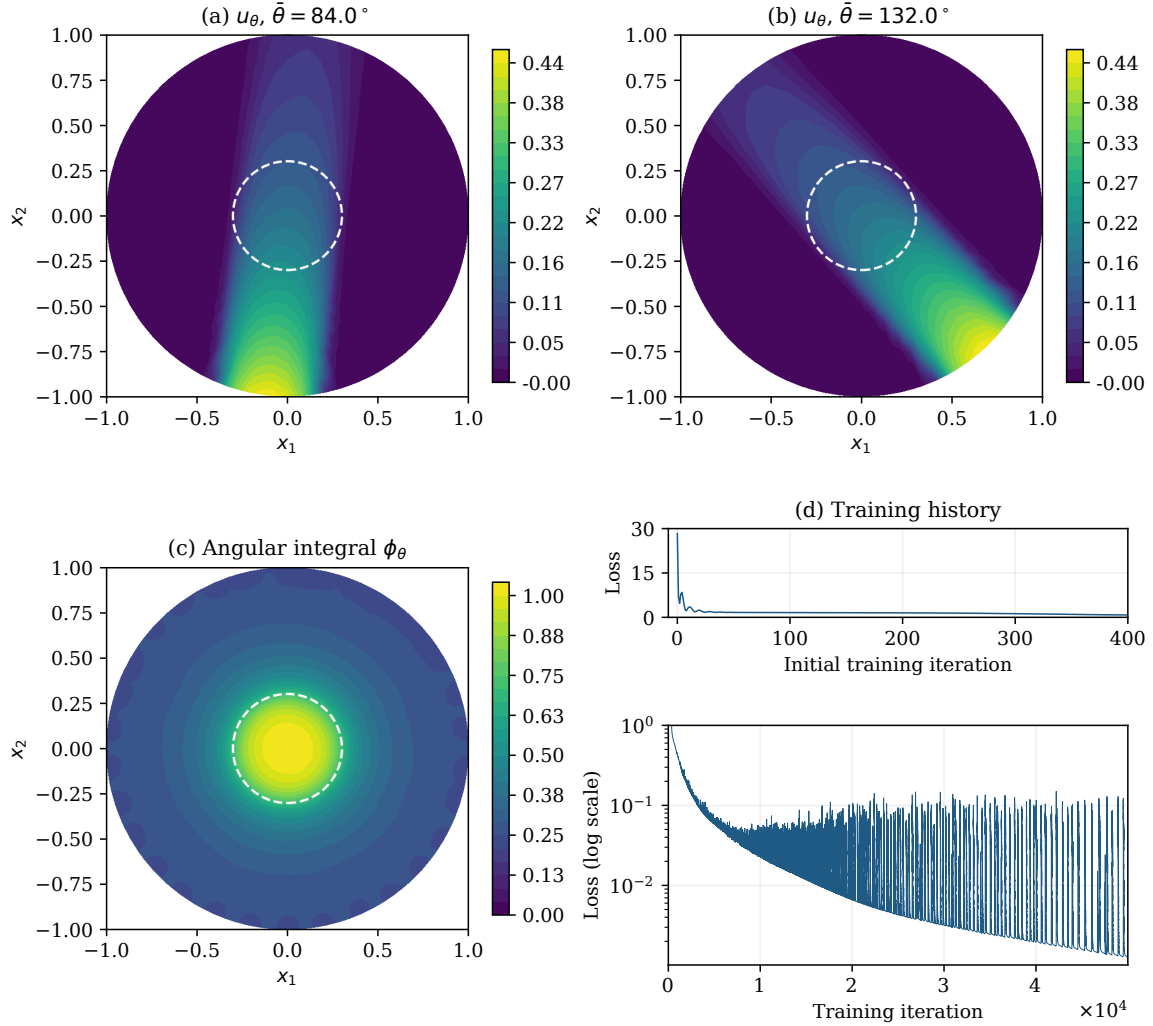


FIGURE 5. Localised target in the unit disc. The directional fields at 84° and 132° , their angular integral and the corresponding training history are taken from the same computation. The dashed circle marks the target boundary. The upper training axis shows the first 400 iterations on a linear scale; the lower axis shows the full history for loss values below 1.

The inflow profile is an unknown in this computation; its spatial and angular dependence is determined together with the state and adjoint. Figure 5 shows attenuating beams passing through the target, with their orientation changing with the incident direction. For this larger attenuation coefficient, the distance travelled before reaching the target has a stronger influence on the optimal inflow through the exponential weight in (20). The control must balance this attenuation against the cost of increasing the incident flux.

The angular integral attains its maximum near the centre and has a gradual transition across the target boundary, with a nonzero contribution outside it. Although X_T is discontinuous, the state must satisfy transport along an entire ray, so the target indicator cannot be reproduced independently at each spatial point. The example therefore examines the directional illumination selected by the constrained problem with the control penalty $\alpha = 10^{-4}$. Direct optimisation of the integrated dose would couple all directions through the adjoint source $D(u) - X_T$. Here the comparison with the characteristic optimal solution distinguishes the effect of transport and regularisation from the network approximation error.

Comparison with the characteristic solution. For the three target geometries, formula (20) provides an independent reference for the optimal state. With constant attenuation, the integrals along each ray are evaluated analytically from its intersection with the target. Table 3 compares the computed state and its angular integral with this reference, using the same spatial and angular quadrature as the computations.

Target	$\ u_\theta - u^*\ _H / \ u^*\ _H$	$\ D(u_\theta) - D(u^*)\ _{L^2(\Omega)} / \ D(u^*)\ _{L^2(\Omega)}$
Circle in the square	1.95×10^{-1}	7.71×10^{-2}
Triangle in the square	7.92×10^{-2}	2.62×10^{-2}
Circle in the disc	5.83×10^{-2}	2.06×10^{-2}

TABLE 3. Relative errors against the characteristic optimal solution for the target examples, evaluated on the training quadrature.

These errors are larger than those in the regular manufactured example, particularly for the small circular target. In all three cases the angular integral has a smaller relative error than the full directional field. Integration combines the directional errors and can reduce their effect on this spatial quantity; accurate angular integrals therefore need not imply equally accurate directional fields. For angular-flux tracking, $D(u^*)$ need not equal X_T : the gradual transition and contribution outside the target in Figure 5 also occur in the characteristic solution. Thus approximation of the optimal field and matching a prescribed integrated dose are distinct questions. The residual values in the different examples should also be read with their loss normalisations and values of α in mind: the stability constant depends on α , so a smaller recorded loss need not correspond to a smaller relative error across these problems.

The boundary norms give the examples complementary roles. The square computations use the stronger unweighted residual, whereas the disc computations use the transport measure. Inclusion (19) explains why both choices control the transport error, and why the unweighted choice also controls the error in U , subject to quadrature control and the fixed loss weights. The paired control diagnostics in Table 2 assess the stronger norm after training with the weaker residual. The square archives provide state fields and loss histories, but not the boundary predictions needed for the analogous control-error comparison. Their state comparisons therefore assess recovery of the respective characteristic optima without providing a quantitative U -error certificate. Since the targets, parameters and geometries also vary, these runs do not isolate the effect of changing the boundary measure.

7.5. Example 7.5: Residual approximation under Monte Carlo quadrature. We now examine the Monte Carlo approximation of the residual functional. This is a separate quadrature experiment, in which the approximating fields remain fixed as the number of samples increases. We perturb the regular manufactured solution to obtain a nonzero residual whose integral can be evaluated exactly. On the unit disc, let (u^*, z^*, g^*) be the solution of Example 7.3, with $\alpha = \sigma = 1$, and set

$$(32) \quad u_\varepsilon = u^* + \varepsilon x \cdot \omega, \quad z_\varepsilon = z^* + \varepsilon |x|^2, \quad g_\varepsilon = g^* + \varepsilon(1 + x_1), \quad \varepsilon = 0.05.$$

Here ε is a fixed perturbation amplitude; it scales the residual functional by ε^2 . The target is unchanged. Writing $d = x \cdot \omega$, the two interior residuals are $\varepsilon(1 + d)$ and $\varepsilon(|x|^2 - 3d)$. On Γ^- , the optimality and inflow residuals are $\varepsilon(1 + x_1 + b)$ and $-\varepsilon(1 + x_1 + b)$, respectively, while the outflow residual is ε . Direct integration therefore gives

$$(33) \quad \mathcal{J}_b(u_\varepsilon, z_\varepsilon, g_\varepsilon) = \frac{\varepsilon^2}{6} (35\pi^2 + 64\pi) \simeq 0.2277075349.$$

For each M , we take M independent uniform samples in $\Omega \times S^1$ and, independently, M samples from the probability measure $(4\pi)^{-1} b dS d\omega$ on Γ^- . The sample means are multiplied by $2\pi^2$ and 4π , respectively. The constant outflow contribution is integrated exactly. We use $M = 2^8, 2^{10}, \dots, 2^{18}$ and repeat the computation independently 256 times. Within each repetition, the smaller sample sets are prefixes of the same unsorted independent sample pools.

Figure 6 shows the root mean square of $\mathcal{J}_{b,M} - \mathcal{J}_b$ over the repetitions. It decreases from 6.89×10^{-3} to 2.28×10^{-4} over the tested range, with a fitted exponent 0.492. This agrees with the $M^{-1/2}$ dependence for fixed fields in (25). Keeping the fields fixed makes this a direct test of Lemma 5.4: the residual density and its variance remain unchanged as M increases. The repeated computations estimate the mean-square sampling error, whose rate need not be visible in a single realisation. The experiment thus isolates the quadrature contribution in the error analysis. During network training the fields depend on the samples, and the uniform estimate of Proposition 5.6 is needed to control that dependence.

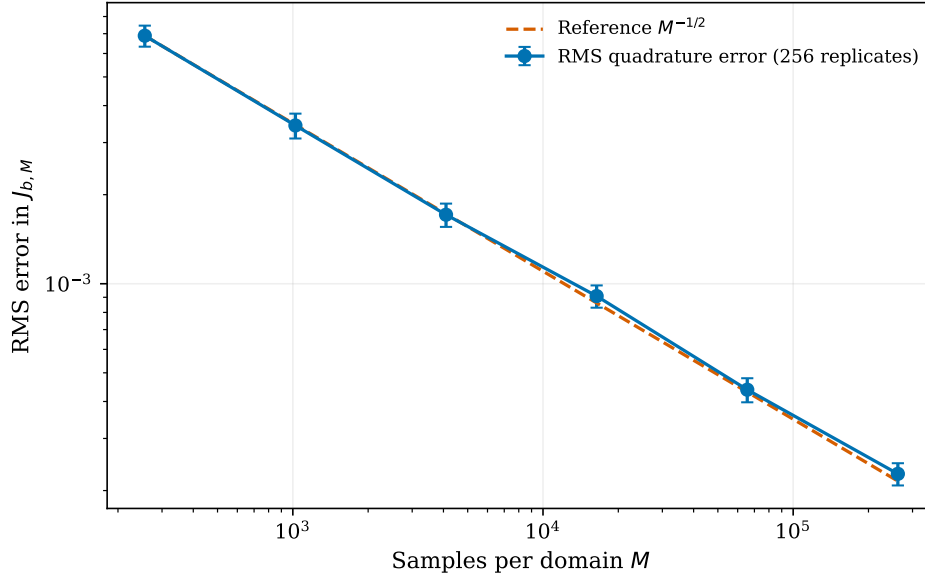


FIGURE 6. Monte Carlo approximation of the residual for the fixed fields (32). The error is computed against the exact value (33). Each point uses 256 independent repetitions; the bars show 1.96 estimated standard errors of the reported root mean square, computed by the delta method. The dashed line has slope $-1/2$. Here M denotes the number of samples in each of the interior and inflow domains.

7.6. Example 7.6: Three-dimensional manufactured benchmark. We finally consider the unit ball in $\mathbb{R}^3$, with directions on S^2 . This example examines the same residual formulation in a five-dimensional phase space. We use the regular manufactured solution of Example 7.3, with $\sigma = \alpha = 1$, so that the computed state, adjoint and inflow control can all be compared with their exact values.

The quadrature uses 30 directions sampled uniformly on S^2 and 10 000 uniformly distributed interior points for each direction, giving 300 000 interior phase-space points. We combine 900 surface points with each direction to obtain 27 000 boundary points. The spatial samples are drawn separately for each direction, whereas the boundary points are reused. All samples remain fixed during training. The network represents the state, adjoint and control jointly and has eight hidden layers of width 64, with tanh activation functions.

Figure 7 shows the normalised residual loss and the relative errors over 2 500 recorded training steps. After the initial decrease, the loss and all three errors fall substantially during the refinement phase. The final relative state, adjoint and control errors are 1.672×10^{-3} , 1.193×10^{-3} and 1.263×10^{-3} , respectively, and the normalised loss is 3.942×10^{-6} . The saved diagnostics remain constant from step 1 826 onwards, so

the extended history also shows the final plateau. These quantities are evaluated on the training samples, using the same normalisations as in the two-dimensional benchmark.

In particular, the reported control error is in G_- . The saved archive contains predictions and azimuths but not the complete direction vectors needed to reconstruct the exact boundary control. An independent unweighted control diagnostic is therefore not reported for this run.

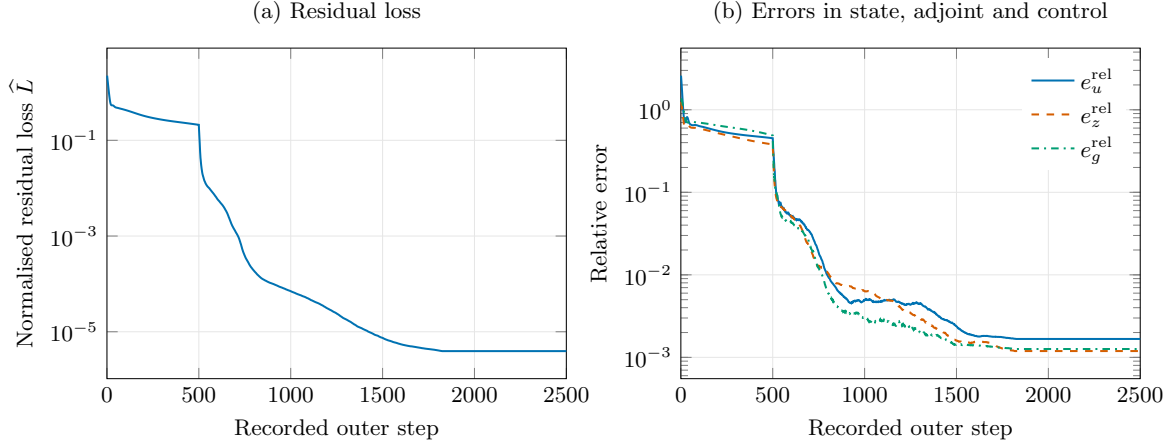


FIGURE 7. Three-dimensional manufactured solution with 10 000 interior points per direction and 30 directions: normalised residual loss (left) and relative state, adjoint and control errors (right). The full history includes the final plateau. All diagnostics use the training samples; a recorded L-BFGS outer step may contain several inner iterations.

The central sections in Figure 8 reveal the directional variation through the interior of the ball. The state is largest towards the incoming part of the boundary and decreases along the transport direction; the negative adjoint tends towards its zero outflow value. Both fields also decay on approaching grazing rays through the factor $A(\Delta)$. The three coordinate sections show how the same characteristic structure appears in different spatial planes, complementing the error histories for the complete state, adjoint and control.

The relative errors remain of order 10^{-3} in this five-dimensional phase-space problem, showing that the coupled residual formulation can be used with a joint neural approximation in three spatial dimensions. The final plateau also indicates that further recorded steps in this run do not improve the reported errors. Since directions are reused across spatial samples, its quadrature has a different sampling structure from the independent phase-space sampling in Example 7.5. The computation provides a three-dimensional manufactured benchmark for the coupled residual formulation. A study of cost, sampling error or convergence with dimension would require controlled refinements.

8. CONCLUSIONS

We have established global residual stability for the KKT system of transport boundary control. The continuum residual bounds the state and adjoint errors in the volume norm and the control error in the flux-weighted boundary norm. The proof reconciles this transport measure with the unweighted control penalty and requires no positive lower bound on $|\omega \cdot n|$. This stability result, followed by the approximation, quadrature and optimisation error analysis, is the main contribution of the paper. Joint neural approximation of the KKT variables provides its numerical realisation. The estimate holds for fixed $\alpha > 0$ and does not assert uniform stability as $\alpha \rightarrow 0$.

The distinction between the two boundary measures also determines what is certified. Strengthening the optimality residual alone gives an estimate in the unweighted control norm and hence a bound on reduced-objective suboptimality. Weighted residual convergence by itself does not imply these stronger conclusions. The postprocessed disc benchmark gives an unweighted relative control error of about 0.10% on the training quadrature.

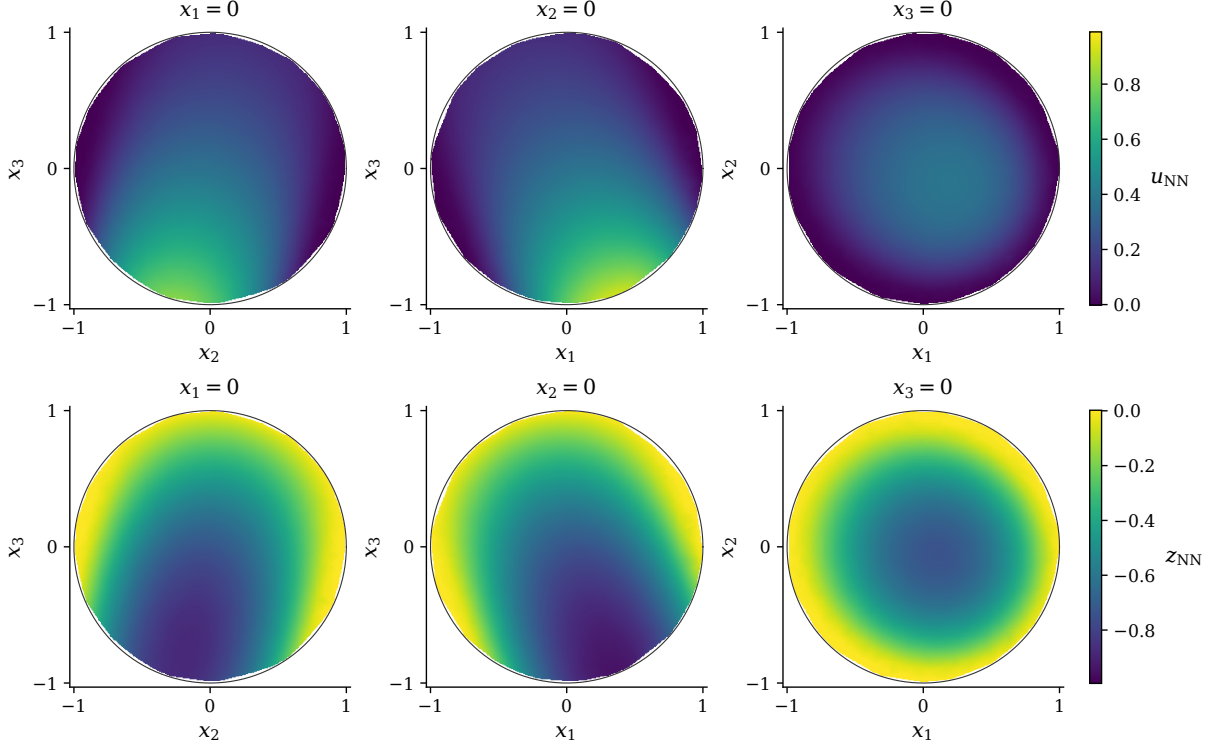


FIGURE 8. Three-dimensional manufactured solution for the first sampled direction. The state (top) and adjoint (bottom) are shown on the central planes $x_1 = 0$, $x_2 = 0$ and $x_3 = 0$. The sections use piecewise-linear interpolation of the 10 000 saved interior values on a tetrahedralisation, with a common colour scale in each row. Regions outside the sampled convex hull are left blank.

At the discrete level, the loss is evaluated by quadrature. For Monte Carlo sampling, we have related the empirical residual to the continuum functional through root-mean-square and high-probability estimates. The fixed-field quadrature error has the $M^{-1/2}$ rate under the stated moment assumptions. For fields selected during training, uniform estimates over the neural class provide the required sampling control. Combining this discrepancy with residual stability yields a probabilistic solution-error bound; combining it with approximation and optimisation gives the convergence result. Deterministic quadrature and independent validation fit the same residual-error argument. The sampling rate concerns the loss; it does not by itself assert the same rate for the solution norm.

The numerical results show how these distinctions enter in practice. The manufactured examples yield recorded relative errors of order 10^{-3} or smaller in two dimensions and order 10^{-3} in three dimensions. For the circular and triangular targets, the characteristic reference reveals larger errors in the directional field, while its angular integral is recovered more accurately. These comparisons assess approximation of the optimal field; they do not identify its angular integral with the prescribed target. The fixed-field Monte Carlo study isolates quadrature error and exhibits the expected root-mean-square rate. The reported neural error norms are evaluated on the training quadrature, so independent spatial and angular validation is a priority for a fuller assessment.

The explicit reliability constants expose another limitation: they may be large at the small regularisation parameters used in target tracking. For angular-flux tracking the characteristic structure improves the bound, but an α^{-2} dependence is unavoidable for the squared-error constant of the unscaled residual. Comparing loss scalings at a fixed geometry, target and approximation class would help assess their effect on training. The present computations use different targets and configurations and do not constitute a controlled numerical study of sensitivity to α .

A direct next step towards treatment planning is to test the angularly coupled dose objective and include nonnegative inflow constraints. The latter leads to a variational inequality and requires a corresponding change to the residual analysis. Scattering introduces further angular coupling, while energy dependence is needed for more complete transport models. For the present model, the characteristic reference gives a useful benchmark for studying refinement, optimiser accuracy and computational cost. The analysis developed here supplies the stability estimate against which such extensions and numerical comparisons can be assessed.

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APPENDIX A. FURTHER ESTIMATES FOR BOUNDARY RESIDUALS

A.1. Explicit stability constants. Using the residual norms from the proof of Theorem 3.2, put $q = \|\mathcal{O}\|^2$ and $R = (R_1, \dots, R_5)^\top$. Define the row vectors

$$(34) \quad \begin{aligned} a_g &= \kappa(qL^{3/2}, \sqrt{L}, 1, qL, 1), \\ a_u &= (L, 0, 0, \sqrt{L}, 0) + \sqrt{L} a_g, \\ a_z &= (0, L, 0, 0, \sqrt{L}) + Lq a_u, \quad \mathbf{M} = \begin{pmatrix} a_u \\ a_z \\ a_g \end{pmatrix}. \end{aligned}$$

The transport estimates in that proof give the componentwise inequality

$$(\|e_u\|_H, \|e_z\|_H, \|e_g\|_{G_-})^\top \leq \mathbf{M}R.$$

Consequently, with $\|\cdot\|_F$ the Frobenius norm,

$$(35) \quad C_{\text{stab}} := 2\|\mathbf{M}\|_F^2 = 2(\|a_u\|_2^2 + \|a_z\|_2^2 + \|a_g\|_2^2)$$

is an admissible constant for (13). The factor two accounts for the factor 1/2 in $\mathcal{J}_b$. The more detailed bound $\|\mathbf{M}R\|_2$ also gives a reliable estimator from the individual continuum residual norms.

For Corollary 3.4, use the same matrix with $\kappa = \alpha^{-1}$ and denote it by $\mathbf{M}_U$. Its proof gives the componentwise estimate with control error in U and $R_U = (\|r_1\|_H, \|r_2\|_H, \|r_3\|_U, \|r_4\|_{G_-}, \|r_5\|_{G_+})^\top$. Thus

$$(36) \quad C_U = 2\|\mathbf{M}_U\|_F^2$$

is sufficient for every bounded observation operator. Its growth is of order α^{-2} for fixed L and $\|\mathcal{O}\|$. For $\mathcal{O} = I$, its values are the raywise columns in Table 4, applied now to $\mathcal{J}_U$ and the error in U .

A.2. Dependence on the regularisation parameter. For $\mathcal{O} = I$, the constant can be sharpened. At an inflow point (y, ω) , let

$$m(y, \omega) = \int_0^{\ell(y, \omega)} \exp\left(-2 \int_0^t \sigma(y + s\omega) ds\right) dt, \quad 0 \leq m \leq L.$$

The characteristic change of variables gives $(Sh, Sk)_H = (mh, k)_{G_-}$ for $h, k \in G_-$. Hence $A = S^*S$ is multiplication by m and commutes with P . It follows that

$$(37) \quad (\alpha I + PA)^{-1}h = \frac{h}{\alpha + bm}, \quad \|(\alpha I + PA)^{-1}\| \leq \alpha^{-1}.$$

Thus one may take $\kappa = \alpha^{-1}$ in (34) for all the angular-flux examples in Section 7. For dose tracking, angular integration couples the directions and the general inverse bound applies.

Some deterioration as $\alpha \rightarrow 0$ is unavoidable. Write $d\mu = dS_x d\omega$ and $A_\varepsilon = \{(x, \omega) \in \Gamma^- : 0 < b < \varepsilon\}$. These sets have positive measure in the present geometries. Choose $h_\varepsilon = \mathbf{1}_{A_\varepsilon} / \|\mathbf{1}_{A_\varepsilon}\|_{G_-}$, and set $e_g = h_\varepsilon$, $e_u = Sh_\varepsilon$, with $T^*e_z = e_u$ and $\gamma^+e_z = 0$. Only $r_3 = (\alpha + bm)h_\varepsilon$ remains nonzero. Since $\|e_g\|_{G_-} = 1$,

$$\mathcal{J}_b \leq \frac{1}{2}(\alpha + \varepsilon L)^2, \quad \|e_u\|_H^2 + \|e_z\|_H^2 + \|e_g\|_{G_-}^2 \geq 1.$$

Letting $\varepsilon \downarrow 0$ shows that every global reliability constant for this unscaled residual satisfies $C_{\text{stab}} \geq 2/\alpha^2$. For fixed L , the sharpened choice (35) has this same order in α , although its numerical value need not be sharp.

α	Square: $L = \sqrt{2}$		Disc or ball: $L = 2$	
	General bound	Raywise bound	General bound	Raywise bound
1	7.176×10^2	1.910×10^2	4.104×10^3	7.120×10^2
10^{-3}	1.731×10^{14}	8.650×10^7	1.409×10^{15}	3.523×10^8
10^{-4}	1.729×10^{18}	8.643×10^9	1.408×10^{19}	3.520×10^{10}

TABLE 4. Rounded values of the explicit choices of C_{stab} from (35) for $\mathcal{O} = I$. The general column uses $\kappa = \alpha^{-1}(1 + L/\alpha)$; the raywise column uses $\kappa = \alpha^{-1}$. These are bounds for the continuum integral residual and the flux-weighted control error, not fitted constants or observed errors.

A.3. Residual weights and reliability. If $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_5)$ with $\lambda_i > 0$ and $\mathcal{J}_\lambda = \frac{1}{2} \sum_i \lambda_i R_i^2$, then

$$\|e_u\|_H^2 + \|e_z\|_H^2 + \|e_g\|_{G_-}^2 \leq 2\|\mathbf{M}\Lambda^{-1/2}\|_2^2 \mathcal{J}_\lambda.$$

For example, replacing r_3 by r_3/α corresponds to $\lambda_3 = \alpha^{-2}$. This preserves the zero-residual solution but changes both the loss and the reliability constant. A smaller constant after rescaling does not by itself give a better error bound or establish improved conditioning of neural optimisation.

For $\mathcal{J}_b$, the constant in Section 5 may be chosen as (35), with the sharper inverse bound of Appendix A.2 for angular-flux tracking. Table 4 shows that this reliable bound can be conservative for small α . The observed ratio $\|V - V^*\|_E^2/\mathcal{J}_b(V)$ does not determine a global constant. For an estimator η of $\|V - V^*\|_E$, the effectivity index is $\eta/\|V - V^*\|_E$; the squared-error/residual ratio is a different quantity. A reverse bound $\mathcal{J}_b(V) \leq C\|V - V^*\|_E^2$ does not hold in general, since the residual contains directional derivatives that the volume error norm does not control.

A.4. Grazing directions and the control norm. Theorem 3.2 does not bound the control error in U . With A_ε and μ as above, now set $h_\varepsilon = \mu(A_\varepsilon)^{-1/2} \mathbf{1}_{A_\varepsilon}$. Then $\|h_\varepsilon\|_U = 1$ and $\|h_\varepsilon\|_{G_-}^2 \leq \varepsilon$. For $(u, z, g) = (u^*, z^*, g^* + h_\varepsilon)$, the residuals are

$$(r_1, r_2, r_3, r_4, r_5) = (0, 0, \alpha h_\varepsilon, -h_\varepsilon, 0), \quad \mathcal{J}_b \leq \frac{1}{2}(1 + \alpha^2)\varepsilon.$$

Thus $\mathcal{J}_b \rightarrow 0$ need not imply $g \rightarrow g^*$ in U . This does not contradict Lemma 3.1: the zero-residual solution is the original minimiser, but convergence of approximate minimisers is certified in the weighted norm.

A.5. Sampling the unweighted optimality residual. For $\mathcal{J}_U$, the measure-based Monte Carlo argument of Section 5.3 applies with $\mu_3 = dS_x d\omega$, leaving the other four measures unchanged. Write $m_-^0 = \int_{\Gamma_-} dS_x d\omega$. For uniform samples Y_j with law $(m_-^0)^{-1} dS_x d\omega$, the optimality term is approximated by

$$\frac{m_-^0}{2M_3} \sum_{j=1}^{M_3} |r_3(V; Y_j)|^2.$$

The same samples may approximate the inflow term by including $b(Y_j)$ in its summand; those estimates do not require independence between different residual terms. If instead the optimality term is sampled from $m_-^{-1} b dS_x d\omega$, its sampled density is $m_- |r_3|^2/(2b)$. Even for bounded r_3 , the second moment of this density can diverge near grazing directions. Finite-variance and bounded-density hypotheses must therefore be verified for the chosen measure. Under the corresponding moment and uniform-class hypotheses, the quadrature proofs in Section 5.3 apply to $\mathcal{J}_U$, and (24) replaces the weighted error bound.

REFERENCES

- [Ash+25] B. S. Ashby et al. “Efficient proton transport modelling for proton beam therapy and biological quantification”. In: *Journal of Mathematical Biology* 90.5, 47 (2025). DOI: 10.1007/s00285-025-02212-1.
- [Bar+26] J. Barry-Straume et al. “Physics-Informed Neural Networks for PDE-Constrained Optimization and Control”. In: *Communications on Applied Mathematics and Computation* 8 (2026), pp. 1283–1306. DOI: 10.1007/s42967-025-00499-x.
- [BFH12] R. Barnard, M. Frank, and M. Herty. “Optimal radiotherapy treatment planning using minimum entropy models”. In: *Applied Mathematics and Computation* 219.5 (2012), pp. 2668–2679. DOI: 10.1016/j.amc.2012.08.099.
- [BG06] P. Bochev and M. D. Gunzburger. “Least-Squares Finite Element Methods for Optimality Systems Arising in Optimization and Control Problems”. In: *SIAM Journal on Numerical Analysis* 43.6 (2006), pp. 2517–2543. DOI: 10.1137/040607848.
- [BG09] P. B. Bochev and M. D. Gunzburger. *Least-Squares Finite Element Methods*. Vol. 166. Applied Mathematical Sciences. Springer, 2009. DOI: 10.1007/b13382.
- [Bor06] T. Bortfeld. “IMRT: a review and preview”. In: *Physics in Medicine and Biology* 51.13 (2006), R363–R379. DOI: 10.1088/0031-9155/51/13/R21.
- [Caf98] R. E. Caflisch. “Monte Carlo and quasi-Monte Carlo methods”. In: *Acta Numerica* 7 (1998), pp. 1–49. DOI: 10.1017/S0962492900002804.
- [Dah+12] W. Dahmen et al. “Adaptive Petrov–Galerkin methods for first order transport equations”. In: *SIAM journal on numerical analysis* 50.5 (2012), pp. 2420–2445.
- [DLM21] T. De Ryck, S. Lanthaler, and S. Mishra. “On the approximation of functions by tanh neural networks”. In: *Neural Networks* 143 (2021), pp. 732–750. DOI: 10.1016/j.neunet.2021.08.015.
- [FHS10] M. Frank, M. Herty, and A. N. Sandjo. “Optimal treatment planning governed by kinetic equations”. In: *Mathematical Models and Methods in Applied Sciences* 20 (2010), pp. 661–678.
- [Hin+09] M. Hinze et al. *Optimization with PDE Constraints*. Vol. 23. Mathematical Modelling: Theory and Applications. Springer, 2009. DOI: 10.1007/978-1-4020-8839-1.
- [Hoe63] W. Hoeffding. “Probability inequalities for sums of bounded random variables”. In: *Journal of the American Statistical Association* 58.301 (1963), pp. 13–30. DOI: 10.1080/01621459.1963.10500830.
- [Hou+24] P. Houston et al. “Efficient high-order space-angle-energy polytopic discontinuous Galerkin finite element methods for linear Boltzmann transport”. In: *Journal of Scientific Computing* 100.2 (2024), p. 52.
- [HSW90] K. Hornik, M. Stinchcombe, and H. White. “Universal approximation of an unknown mapping and its derivatives using multilayer feedforward networks”. In: *Neural Networks* 3.5 (1990), pp. 551–560. DOI: 10.1016/0893-6080(90)90005-6.
- [KB15] D. P. Kingma and J. Ba. “Adam: A Method for Stochastic Optimization”. In: *International Conference on Learning Representations*. 2015.
- [LN89] D. C. Liu and J. Nocedal. “On the limited memory BFGS method for large scale optimization”. In: *Mathematical Programming* 45 (1989), pp. 503–528. DOI: 10.1007/BF01589116.
- [Lu+21] L. Lu et al. “Physics-informed neural networks with hard constraints for inverse design”. In: *SIAM Journal on Scientific Computing* 43.6 (2021), B1105–B1132.
- [LZ20] Q. Liu and S. Zhang. “Adaptive least-squares finite element methods for linear transport equations based on an $H(\text{div})$ flux reformulation”. In: *Computer Methods in Applied Mechanics and Engineering* 366 (2020), p. 113041. DOI: 10.1016/j.cma.2020.113041.
- [Mak+25] C. Makridakis, A. Pim, T. Pryer, and N. Rekatsinas. *Deep Uzawa for Kinetic Transport with Lagrange-Enforced Boundaries*. Preprint. 2025. DOI: 10.48550/arXiv.2507.19907.
- [MM21] S. Mishra and R. Molinaro. “Physics informed neural networks for simulating radiative transfer”. In: *Journal of Quantitative Spectroscopy and Radiative Transfer* 270 (2021), p. 107705.
- [MM23] S. Mishra and R. Molinaro. “Estimates on the generalization error of physics-informed neural networks for approximating PDEs”. In: *IMA Journal of Numerical Analysis* 43.1 (2023), pp. 1–43. DOI: 10.1093/imanum/drab093.

- [MN23] S. Mowlavi and S. Nabi. “Optimal control of PDEs using physics-informed neural networks”. In: *Journal of Computational Physics* 473 (2023), p. 111731. DOI: 10.1016/j.jcp.2022.111731.
- [MPP24] C. G. Makridakis, A. Pim, and T. Pryer. “Deep Uzawa for PDE constrained optimisation”. In: *arXiv preprint arXiv:2410.17359* (2024).
- [MPP25] C. G. Makridakis, A. Pim, and T. Pryer. “A Deep Uzawa-Lagrange Multiplier Approach for Boundary Conditions in PINNs and Deep Ritz Methods”. In: *Journal of Machine Learning* 4.3 (2025), pp. 166–191. DOI: 10.4208/jml.250107.
- [PP25] J. Power and T. Pryer. “Adaptive regularisation for PDE-constrained optimal control”. In: *Journal of Computational and Applied Mathematics* 470, 116651 (2025). DOI: 10.1016/j.cam.2025.116651.
- [RPK19] M. Raissi, P. Perdikaris, and G. E. Karniadakis. “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations”. In: *Journal of Computational Physics* 378 (2019), pp. 686–707.
- [SZK23] Y. Shin, Z. Zhang, and G. E. Karniadakis. “Error estimates of residual minimization using neural networks for linear PDEs”. In: *Journal of Machine Learning for Modeling and Computing* 4.4 (2023), pp. 73–101. DOI: 10.1615/JMachLearnModelComput.2023050411.
- [Tre19] L. N. Trefethen. *Approximation Theory and Approximation Practice, Extended Edition*. Society for Industrial and Applied Mathematics, 2019. DOI: 10.1137/1.9781611975949.
- [Zha+26] T. Zhang et al. “Enhanced physics-informed neural networks for PDE-constrained optimal control: A synergistic approach with adversarial attack and scale adjustment”. In: *Physica A: Statistical Mechanics and its Applications* 682 (2026), p. 131201. DOI: 10.1016/j.physa.2025.131201.

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